

Help Converts Newcomers, Not Veterans: Generalized Reciprocity and Platform Engagement on Stack Overflow

Abstract

Whether receiving help on an online platform increases subsequent helping is an open empirical question. Prior observational studies report mixed results, and designs that do not net out baseline user activity may overstate reciprocity. We develop a matched difference-in-differences survival analysis of over 21 million Stack Overflow questions, comparing users whose questions received an answer with propensity-score-matched users whose questions did not. Receiving an answer is associated with a small increase in the hazard of subsequently answering other users' questions (difference-in-differences increment at answer arrival, hazard ratio $[HR] = 1.03$, 95% CI $[1.02, 1.05]$), and this association is steeply graded by user tenure: it is strongest among newcomers ($HR = [AO:nc-arrival-hr]$ in the first week) and fades to zero among the longest-tenured users. The newcomer gradient is the most robust pattern in the data: the first-week association also holds on the pre-2020, pre-generative-AI subsample. Faster answers are associated with a larger arrival increment, with a pronounced early peak for answers arriving within 30–60 minutes — qualified support for an immediacy account, though we cannot fully separate answer speed from answer quality. The data identify short-run behavioral associations, not the psychological mechanisms (gratitude, moral obligation, incentive displacement) that theory proposes. The pattern is consistent with generalized reciprocity operating primarily as a contributor-recruitment channel at the point of entry, before platform-specific incentives take hold, and it suggests newcomer-focused design implications.

Keywords: generalized reciprocity, online communities, knowledge sharing, survival analysis, platform design, Stack Overflow

1. Introduction

Whether receiving help on an online platform motivates subsequent helping behavior has been a persistent question in the study of online communities. Generalized reciprocity, the tendency to help others after being helped oneself (Nowak and Roch, 2007; Baker and Bulkley, 2014), is among the mechanisms proposed to sustain cooperation on platforms like Stack Overflow and Wikipedia, where direct exchange between the same two individuals is rare. If receiving help reliably increases the probability of giving it, cooperation may sustain itself through cascading helping acts, even among strangers (Fowler and Christakis, 2010). The behavioral evidence, however, is inconsistent: some observational studies find substantial reciprocity effects (Baker and Bulkley, 2014; Chen et al., 2019), while others find weak or null relationships (Wasko and Faraj, 2005; Joyce and Kraut, 2006). We argue that much of this inconsistency is methodological. Our contribution is a better-identified short-run behavioral test on one platform: whether helping rises after answer receipt. The data do not directly measure the psychological mechanisms that theory proposes (gratitude, moral obligation, incentive displacement); we test only the behavioral consequence those mechanisms predict.

Observational studies of reciprocity on platforms face a persistent confound: more engaged users are more likely both to seek help and to provide it, regardless of whether their own questions are resolved. Studies that correlate help received with help given, without accounting for this baseline activity, conflate reciprocity with general user engagement (Baker and Bulkley, 2014; Chen et al., 2018; Joyce and Kraut, 2006). A related limitation is that prior designs have largely treated receiving help as a one-time event, even though help-seeking and helping occur continuously.

We address both problems by developing a matched difference-in-differences (DiD) survival analysis, applied to over 21 million questions from Stack Overflow. For each question a user posts, we define observation windows before and after the question and compare changes in helping behavior between users whose question received an answer (treated) and propensity-score-matched users whose question did not (control). We use Cox proportional hazards models to account for the continuous-time dynamics of helping behavior and the staggered, time-varying structure of treatment.

We develop two theoretical predictions alongside the basic reciprocity hypothesis. Drawing on theories of socialization and norm internalization, we

predict that the reciprocity effect is stronger among newcomers than among experienced users (H2). Newcomers have not yet internalized community-specific norms or learned to value platform incentives such as reputation points; general moral obligations to reciprocate are therefore more likely to drive their behavior. As users accumulate experience, these general-purpose triggers may be displaced by community-specific motivations. We also predict that faster responses amplify the reciprocity effect (H3), because gratitude is more intense when help arrives while the need is still salient (McCullough et al., 2001; Tsang, 2006).

The results support the baseline reciprocity hypothesis (H1) in a modest, newcomer-concentrated form, and offer qualified support for the tenure gradient (H2) and the immediacy pattern (H3). Relative to matched controls and to their own pre-question baseline, users who receive an answer show a small elevation in their subsequent answering of other users' questions: the pooled difference-in-differences increment at answer arrival is $HR = 1.03$ (95% CI [1.02, 1.05], $p < .001$). The association is steeply graded by tenure: the arrival increment is largest among newcomers ($HR = [AO:nc-arrival-hr]$ in the first week, $p < .001$) and fades to zero among the longest-tenured users. Our estimand throughout is the increment at answer arrival (β_4), the difference-in-differences contrast that nets out any treated-control divergence arising before the answer; we report the waiting-period term (β_2) alongside it as an explicit parallel-trends diagnostic rather than folding it into the effect, and we re-estimate the model under broader outcome definitions that add lower-cost helping acts (comments, edits) in a decomposition (Table 6) to show where the signal resides.

Our study makes three contributions. First, using a design built to remove the activity confound that inflates observational estimates, we provide a better-identified short-run behavioral test: receiving an answer is associated with a small increase in subsequent helping on Stack Overflow (arrival-increment $HR = 1.03$), substantially smaller than prior observational reports and $[AO:pooled-quality-direction - attenuated/reversed]$ once answer-quality controls are added. Second, we show that this association is steeply concentrated among newcomers and absent among veterans — the tenure gradient is the most robust and informative pattern in the data, though we cannot cleanly separate reciprocity from selection into being answered. Third, methodologically, we decompose the outcome by help type and enter the waiting period as an explicit term, making pre-answer divergence visible rather than silently absorbed into the treatment effect.

2. Theory

2.1. Generalized Reciprocity

A central mechanism proposed to support cooperation in online communities is *generalized reciprocity*, the tendency for individuals who have received help to help others in turn, even when the recipient is not the original benefactor (Nowak and Roch, 2007; Fowler and Christakis, 2010). Unlike direct reciprocity, it requires no repeated exchange with the same person: it operates as a pay-it-forward principle, sustaining collective contributions in groups — such as online knowledge-sharing platforms — where interactions are infrequent and non-recurring.

Socio-psychological research identifies two complementary mechanisms underlying generalized reciprocity: moral reasoning and emotional response. On the moral side, individuals who reason by universalization — asking what would happen if all members requested more help than they gave — conclude that widespread free-riding would undermine the community as a source of help, generating a shared obligation to reciprocate toward the group (Levine et al., 2020; Curry et al., 2019). On the emotional side, gratitude aroused by the perceived generosity of others motivates prosocial action beyond direct reciprocation (McCullough et al., 2001; Tsang, 2006; Bartlett and DeSteno, 2006) and is especially relevant where free riding is easy, as on knowledge-sharing platforms (Molm et al., 2007; Elster, 2009). These moral and emotional pathways are the *proposed* generative mechanisms behind generalized reciprocity; our design tests only whether helping rises after answer receipt, not whether gratitude or moral obligation is experienced.

Evidence that generalized reciprocity can operate — and can initiate cascading acts of generosity (Fowler and Christakis, 2010; Boyd and Richerson, 1989) — is robust: comparative research indicates that the underlying norm is globally shared (Curry et al., 2019), and experimental demonstrations span laboratory (Greiner and Levati, 2005; Molm et al., 2007; Stanca, 2009) and field settings (Mujcic and Leibbrandt, 2018; Fowler and Christakis, 2010). The open question is how strongly the mechanism operates in situ on large platforms — a question observational data have answered inconsistently, for reasons we argue are largely methodological.

2.2. Empirical Challenges in Measuring Generalized Reciprocity

One challenge is the reliance on self-reported motives in surveys and interviews. Contributors frequently describe reciprocity as a central reason for

their engagement (Wasko and Faraj, 2000; Lakhani, 2003), yet people have difficulty articulating the moral motives behind their actions (Vaisey, 2009; McClelland et al., 1989), and introspection struggles to separate reciprocity from altruism, conformity to perceived norms, status pursuit, or the popular rhetoric of “giving back” as post-hoc rationalization (Batson, 1991; Bénabou and Tirole, 2006). Self-reported data may therefore overstate the prevalence of reciprocity as a discrete motive.

The more fundamental challenge is the endogeneity of help-seeking and help-giving in observational data. Users who ask questions on platforms like Stack Overflow are also disproportionately likely to answer questions — not because receiving help triggered a reciprocal response, but because both behaviors reflect the same underlying characteristics, such as greater platform engagement. Studies that correlate help received with help given without accounting for this baseline activity effectively confound reciprocity with user type (Baker and Bulkley, 2014; Chen et al., 2018; Joyce and Kraut, 2006). This confound is the central identification challenge that motivates the matched difference-in-differences design we develop in Section 4.

Prior observational studies of online reciprocity report mixed magnitudes. Baker and Bulkley (2014) and Chen et al. (2019) find positive associations between help received and subsequent helping; Wasko and Faraj (2005) and Joyce and Kraut (2006) find weak or null relationships. Our matched DiD estimates sit below the larger positive reports (pooled arrival-increment $HR = 1.03$; newcomer arrival increment $\approx [[AO:nc-arrival-hr]]$), consistent with the view that designs that do not net out baseline activity overstate the effect. The tenure gradient — clear elevation among newcomers, null among veterans — offers one reconciliation of the mixed prior findings.

2.3. Reciprocity and the User Trajectory

We propose that the mixed findings in the literature can be partly reconciled by attending to a dimension that prior research has largely overlooked: the user trajectory from newcomer to experienced contributor. We argue that generalized reciprocity is strongest for new users and attenuates as users gain experience on the platform.

The logic of this prediction draws on theories of socialization and norm internalization. When a user first encounters an online community, they are unfamiliar with its specific norms, social structures, and reward systems. At this stage, community-specific incentives, such as reputation points, badges, or leaderboard positions, carry little motivational weight because the

newcomer has not yet learned to value them. Instead, initial contributions are guided by more general moral principles that operate independently of community-specific knowledge. Generalized reciprocity is precisely such a principle: it requires only that the individual has received help and feels a moral obligation or emotional impulse to help in return, without any need to understand or identify with the particular community (Levine et al., 2020; Curry et al., 2019).

As users interact with the community, they gradually learn its norms and develop a sense of membership. On platforms like Stack Overflow, these norms are codified in explicit reputation systems that reward helpful behavior with visible status markers. Once users internalize these community-specific norms, the broad moral obligation of reciprocity may be displaced by more localized incentive structures — a theoretical prediction our design cannot adjudicate directly, because we do not observe reputation salience or user motives. The logic of universalization (“I should help because I was helped”) gives way to the logic of community membership (“I should help because that is what valued members do”). This norm displacement has been theorized in related contexts by Bénabou and Tirole (2006), who show how external incentives can crowd out moral motives.

Two further mechanisms accelerate this decline. As users accumulate experiences of receiving help, each additional instance becomes less surprising, and the gratitude that fuels reciprocity habituates (McCullough et al., 2008). And as users learn that the platform rewards helping with reputation and status, they may increasingly attribute others’ helpfulness to status pursuit rather than genuine generosity, further dampening that gratitude.

This account predicts that generalized reciprocity is strongest among newcomers and declines with experience. This also helps explain why prior studies have reached different conclusions about the importance of reciprocity: those that focus on or oversample experienced users may underestimate reciprocity, while those that capture newcomer behavior may find stronger effects.

HYPOTHESIS 1 (H1). Receiving help increases a user’s propensity to help others (generalized reciprocity effect).

HYPOTHESIS 2 (H2). The generalized reciprocity effect is strongest among newcomers and decreases as users gain experience on the platform.

The stakes of this prediction extend beyond reconciling mixed findings. If the reciprocity association is concentrated among newcomers, generalized reciprocity would function less as a general engine of cooperation among established members than as a *contributor-recruitment* channel operating at the point of entry, before community-specific incentives take hold — a design implication we develop in Section 6.4.

2.4. Response Time and the Immediacy of Reciprocity

The emotional mechanisms underlying reciprocity suggest that the speed of help matters for its strength. Gratitude, the primary proposed emotional driver of generalized reciprocity (McCullough et al., 2001), is triggered by the perception of a benefactor’s generosity, and its intensity is likely to depend on the characteristics of the helping experience — including its timeliness.

A swift response can amplify the experience of receiving help: a fast answer signals that the community is attentive and generous, and it arrives while the asker’s need is still salient, when gratitude should be most intense. Conversely, when an answer arrives hours or days later, the original need may have faded or been met elsewhere, the associative link between asking and receiving weakens, and the delay itself may signal an unresponsive community — attenuating the emotional and moral responses that underpin reciprocation.

These considerations lead to a straightforward prediction:

HYPOTHESIS 3 (H3). *Faster responses to a user’s question strengthen the generalized reciprocity effect, while slower responses weaken it.*

We note that this hypothesis follows from the dominant theoretical accounts of generalized reciprocity, which emphasize emotional immediacy and the salience of the helping experience. However, as we will show, field settings introduce structural dynamics, particularly the behavioral consequences of users being online and offline, that qualify the predictions of laboratory-based theory. We return to this issue when presenting our results.

3. Empirical Setting: Stack Overflow

Stack Overflow is well suited to examining generalized reciprocity in online knowledge sharing. The platform provides publicly available quarterly data dumps containing longitudinal records of user behavior, including knowledge-seeking and knowledge-providing activities dating back to the

platform’s inception. These data allow us to reconstruct helping patterns over time and to analyze the temporal sequence of receiving and providing help on which any study of generalized reciprocity depends. Stack Overflow’s active question-and-answer volume has declined sharply since the release of generative AI tools in late 2022; our sample spans platform inception through 1 April 2025 and therefore captures a period that includes but predates the steepest LLM-driven decline. We treat the findings as a historical snapshot of one large, technical, relatively anonymous, status-driven platform; external validity to other communities is discussed in Section 6.

At its core, Stack Overflow is a question-and-answer platform for software developers and technologists. As of mid-2025, the site had accumulated over 24 million questions and 36 million answers from more than 4.5 million users. Its user base spans a wide range of experience levels, from learners to professional software developers (Stack Overflow, 2024). Questions on the platform must be tightly focused on specific programming problems; broad or opinion-based inquiries are typically closed by community moderation (Stack Exchange Inc., 2024). The most frequently discussed topics, as reflected in question tags, include JavaScript, Python, Java, C#, PHP, Android, and HTML, spanning areas from web development and mobile applications to data science and machine learning. Stack Overflow is built around several core features: tags categorize questions by topic, facilitating expertise matching; a voting system allows users to upvote or downvote questions and answers to assess quality; and reputation points are earned for valued contributions such as providing high-quality answers (Stack Exchange Inc., 2024).

For our study, the key feature is the question-and-answer interaction between users. When a user posts a question, other users may provide answers. This creates a timestamped sequence (question posted, answer received) that we exploit to measure the temporal dynamics of generalized reciprocity. On Stack Overflow, question-askers and answerers are typically different individuals, which makes the platform well suited to studying *generalized* reciprocity: users who receive help from one community member may subsequently help entirely different members.

4. Data and Methods

To test our predictions, we develop a matched staggered difference-in-differences survival analysis that leverages the temporal structure of help-seeking and help-giving on Stack Overflow. Our design compares changes

in helping behavior (measured as the hazard of posting an answer) before and after a user receives an answer, between users who receive an answer and matched users who do not. This approach integrates two methodological traditions: the staggered difference-in-differences framework of Callaway and Sant’Anna (2021), which accommodates variation in treatment timing and treatment effect heterogeneity, and the counting-process formulation of the Cox proportional hazards model (Andersen and Gill, 1982), which allows treatment status and other covariates to vary continuously over the observation window.

4.1. Study Design

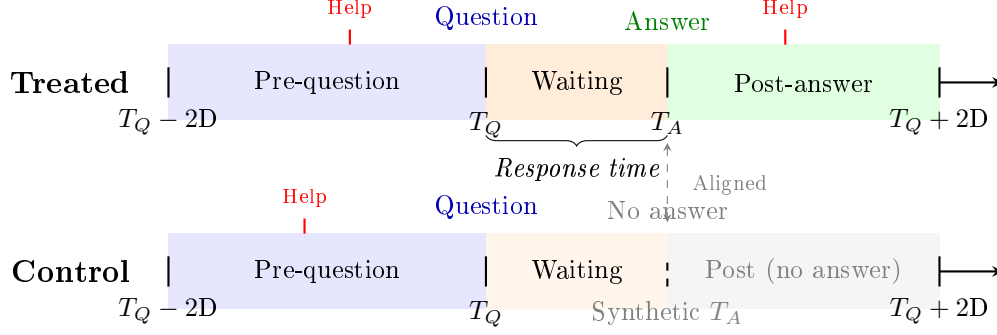
For each question posted by a user, we construct a timeline centered on two key events: the posting of the question (T_{question}) and, for questions that receive an answer, the time at which the first answer with a non-negative vote score is posted (T_{answer}). We define an observation window spanning the two days before the question is posted ($T_{\text{question}} - 2\text{D}$) through the two days after ($T_{\text{question}} + 2\text{D}$). This yields a 4-day timeline for each question, divided into distinct phases.

We define three phases within each observation window (see Figure 1). The first phase, the *pre-question phase*, is from the start of the observation window to the posting of the question ($T_{\text{question}} - 2\text{D}$ to T_{question}). This phase captures baseline helping activity before the focal question is asked. The second phase, the *waiting period*, is from the posting of the question to the receipt of the first answer (T_{question} to T_{answer}). During this interval, the user has asked for help but has not yet received it. The final *post-answer phase* is from the receipt of the answer to the end of the observation window (T_{answer} to $T_{\text{question}} + 2\text{D}$). This is the phase in which the reciprocity effect, if it exists, should manifest.

A key feature of our design is that the post-answer (treated) phase begins at different elapsed times across observations, depending on how quickly the posted question receives a response.

This staggered timing of the post-answer phase corresponds to the setting studied by Callaway and Sant’Anna (2021), who show that standard two-way fixed-effects estimators can be biased when treatment timing varies, and treatment effects are heterogeneous. Our approach avoids this pitfall by explicitly modeling the time-varying change in treatment status within each observation window and by comparing treated observations only to not-yet-treated or never-treated controls.

Figure 1: Study Design: Matched Difference-in-Differences Survival Analysis



Note. Each observation window spans 4 days centered on the focal question (T_Q). The treated timeline (top) shows three phases: the pre-question phase (baseline), the waiting period (question posted, no answer yet), and the post-answer phase (after the first qualifying answer T_A arrives). The control timeline (bottom) is from a propensity-score-matched user whose question did not receive an answer. To ensure temporal alignment, the control’s post-question period is split at a synthetic transition point matching the treated user’s response time. Red markers indicate help events (answers posted by the focal user to other users’ questions). The treatment effect is the differential change in helping hazard between treated and control observations in the post-answer phase, relative to the pre-question baseline. The staggered timing of T_A across observations motivates our use of matched DiD.

For control users (those whose questions did not receive an answer), there is no natural transition between the waiting period and the post-answer phase. To maintain temporal alignment with the matched treated observation, we assign the control observation a synthetic phase transition at the same elapsed time as the treated user’s response time. That is, if the treated user’s answer was posted 4 hours after their question, the matched control observation’s timeline is split at $T_{\text{question}} + 4\text{h}$, creating an aligned “post” phase of equal duration. This ensures that any differences between treated and control observations in the post-answer phase are not driven by differences in the length of the observation period.

The survival clock in our Cox models is centered on posting of the respective question: time zero corresponds to the start of the observation window ($T_{\text{question}} - 2D$), so that the question itself is posted at day 2 of the timeline and the observation window closes at day 4. All help events (answers posted by the focal user) are recorded with their exact timestamps and converted to elapsed time on this clock.

Within each timeline, we track all instances in which the user provides an answer to another user’s question. These helping events constitute the outcome in our survival analysis; broader outcome definitions that also count comments and edits are examined in the decomposition (Table 6). Conceptually, our design asks: Does the hazard of helping increase after a user receives an answer, relative to the change in hazard for a matched user who does not receive an answer? We present descriptive evidence on the raw help rates over this timeline in Section 5 (Figure 2), before turning to the formal Cox model estimates.

4.2. Data

We used the full sample of non-deleted questions from Stack Overflow’s inception on 31 July 2008 through 1 April 2025. We addressed two edge cases that would render our fixed-length observation window invalid. First, for early platform questions, the pre-period could predate the platform’s launch, meaning no answers were possible at that time. We removed 1,080 such cases. Second, due to the data dump’s 1 April 2025 cutoff, the post-period could be prematurely truncated. We excluded 8,029 cases affected by this. We then excluded 2,671,121 questions that received an answer from the question poster. These filtering steps resulted in a pre-matching dataset of 21,033,429 questions from 4,764,048 unique users. After propensity score matching (described below), the matched file comprises 24,349,760 question-level rows (12,174,880 matched pairs; some control questions appear more than once) from 3,732,706 unique users, with a 50% treatment rate by construction. All matched rows enter the Cox analysis; the helping outcome is the posting of an answer to another user’s question (self-directed accepts excluded), yielding 3,435,973 helping events in the pooled analysis sample (Table 5).

4.3. Variables

Table 1 provides an overview of the variables used in our analysis. The phase terminology is consistent with Figure 1: the pre-question phase captures baseline behavior, the waiting period spans the interval from question to answer, and the post-answer phase captures the period after the answer arrives. We use “post-question” to refer to the combined waiting period and post-answer phase whenever distinguishing between these two sub-periods is not necessary.

Table 1: Overview of Variables

Variable	Description
<i>Outcome (Survival Event)</i>	
helpEvent	A binary event indicator equal to 1 at times when the user posts an answer to another user’s question within the observation window (self-directed accepts excluded). Broader definitions adding comments and edits are reported in the outcome decomposition.
<i>Time-Varying Covariates</i>	
PostQuestion	Binary; 1 for intervals occurring after the user posts their question ($t \geq T_{\text{question}}$), 0 otherwise. Captures the combined post-question period (waiting period + post-answer phase).
PostAnswer	Binary; 1 for intervals occurring after the user receives their first answer ($t \geq T_{\text{answer}}$), 0 otherwise. For control users (no answer received), this variable switches on at the synthetic transition point at the matched treated user’s response time. Isolates the post-answer phase.
treatedPostQuestion	Interaction of treatment $\times$ phasePostQuestion; captures differential behavior of treated users in the entire period after the question.
isTreatedActive	Interaction of treatment $\times$ phasePostAnswer; the primary treatment effect capturing the additional lift in helping hazard after receiving an answer, beyond any change attributable to the act of asking a question.
ResponseTime	Time between question and the first answer (excluding down-voted answers) of the treated question. We include interactions with all four variables above.

Outcome.. The event of interest is the posting of an answer to another user’s question — the platform’s central helping act. Within each window we track when (and whether) the focal user answers any other user’s question, and the Cox model estimates the instantaneous hazard of this helping event as a function of time-varying covariates that capture the phase of the observation timeline and the treatment status. Self-directed acts (accepting an answer on one’s own question) are excluded. To address the concern that answers alone may understate lower-cost reciprocity, we re-estimate the pooled model under broader outcome definitions — comments only, edits only, and composites that add these channels to answers — and report the full decomposition in Table 6.

Why answers anchor the construct.. We treat the posting of an answer as the theoretically central helping act. Generalized reciprocity, as modeled in the experimental literature, is a *costly* prosocial contribution: individuals pass a benefit to an unrelated third party at real expense to themselves (Greiner and Levati, 2005; Stanca, 2009; Mujic and Leibbrandt, 2018), and it is the willingness to bear that cost — not mere approval or endorsement — that distinguishes reciprocity from low-cost signaling (Molm et al., 2007). Answering a stranger’s question is the platform analogue: it demands substantive effort and yields a public good for the community. Lower-cost gestures such as comments, upvotes, and edits index engagement, curation, or approval more than the effortful contribution the construct denotes. We therefore anchor the outcome on answers and report comment- and edit-inclusive alternatives in the decomposition (Table 6); upvotes, the one lower-cost channel we cannot observe at all, lie outside our processed data. As the decomposition shows, this choice does not manufacture the effect: comments alone carry no elevation, so broadening the outcome dilutes rather than inflates the estimate (see Section 6.5).

Treatment.. The treatment is the receipt of an answer. We operationalize this using *treatment*, a binary variable indicating whether the user’s question received at least one answer with a non-negative score. We adopt this operationalization rather than Stack Overflow’s “accepted answer” mechanism to address endogeneity: marking an answer as accepted requires the user to return to the site, which may itself correlate with helping behavior.

Response Time.. For each treated observation, the response-time variable used in the interaction terms is $\log(1 + t_{\text{answer}})$, where t_{answer} is the elapsed

time in hours from the start of the observation window ($T_Q - 2D$) to the arrival of the first answer with a non-negative vote score. Because $T_Q - 2D$ is a fixed origin common to all observations, this is a monotone transformation of the actual response time in hours ($t_{\text{answer}} - t_{\text{question}}$) and preserves the full ordering of fast versus slow answers; however, it shifts the scale by the constant pre-question window length. For numerical stability, both response-time interaction terms ($\text{isTreatedActive} \times \log(1 + t_{\text{answer}})$ and $\text{treatedPostQuestion} \times \log(1 + t_{\text{answer}})$) are clipped to their 5th–95th percentile range and then standardized (z-scored) before model fitting. Consequently, the reported interaction coefficients γ and δ are expressed per one standard deviation of the respective interaction term rather than per unit of $\log(1 + t_{\text{answer}})$.

User Experience. To test Hypothesis 2, we stratify our analysis by user tenure, defined as the number of days between a user’s first activity on the platform and the start of the observation window. We define seven tenure buckets using right-closed intervals: $[0, 7]$, $(7, 30]$, $(30, 180]$, $(180, 365]$, $(365, 1095]$, $(1095, 2190]$, and $(2190, \infty)$ days, corresponding to the labels less than one week, one week to one month, one to six months, six to twelve months, one to three years, three to six years, and more than six years, respectively. Running separate models by bucket enables us to examine how the reciprocity effect varies across the user trajectory without imposing a parametric functional form.

4.4. Propensity Score Matching

Because the receipt of an answer (the treatment) is correlated with user characteristics (more experienced users, for instance, might tend to ask higher-quality questions that are more likely to receive answers), we employ propensity score matching (Rosenbaum and Rubin, 1983) to improve the comparability of the treatment and control groups. Importantly, matching is performed *across different users*: each treated question (from a user who received an answer) is matched to a control question (from a different user whose question did not receive an answer) that has a similar propensity to receive an answer based on observable pre-treatment characteristics. This cross-user matching ensures that treated and control observations are independent conditional on the matched covariates, while the DiD structure within each pair absorbs residual differences.

We estimate the propensity score using pre-treatment covariates that jointly capture user experience, recent activity, and question characteristics, as listed in Table 2. The covariate *tagAcceptShareAvg* addresses the concern that question complexity may confound the treatment: questions in tags with low accepted-answer rates are inherently more difficult, and matching on this rate ensures that treated and control questions come from similarly tractable topic areas. Matching on the tag-level accepted-answer rate also matters for the response-time analysis, because it mitigates the concern that slower response times simply proxy for differently sized sub-communities with lower throughput. We use nearest-neighbor matching *with replacement* and a caliper of 0.05 on the propensity score, requiring in addition an exact match on the calendar year, the question’s primary tag, and the number of tags. Because the pool of unanswered questions is smaller than the pool of answered questions, some control questions are matched to more than one treated question; the matched file therefore contains more rows than unique question pairs (12.2 million pairs, 24.3 million rows). Each row enters the Cox analysis as an independent observation window; standard errors are discussed in the robustness appendix.

Table 3 displays the covariate balance before and after matching. Before matching, several covariates showed a meaningful imbalance between the treated and control groups. For instance, treated users had asked more questions in the prior 30 days (SMD = 0.197) and 7 days (SMD = 0.174), reflecting that users who ask questions that attract answers tend to be more recently active; their questions were also longer and posted by higher-reputation askers (body length SMD = −0.183; owner reputation SMD = 0.115). The largest imbalance was in the tag-level accepted-answer rate (SMD = 0.695), confirming that question topic characteristics strongly predict answer receipt and underscoring the importance of matching on this variable. To address the observable-selection concern directly, the propensity model also includes posting hour and day, question and title length, and owner reputation, and treated and control questions are matched exactly on the number of tags in addition to the primary tag. After matching, all standardized mean differences fell well below the conventional threshold of 0.1, with the largest residual imbalance below 0.05 (see Table 3). This confirms adequate balance across all matching covariates; the full set, including posting-time features, is illustrated in the Love plot in Appendix Appendix B.

Table 2: Definition of Matching Covariates

Variable name	Definition
<i>Exact matching</i>	
calendarYear	Calendar year exact matching indicating the year of the observation, controlling for secular trends in platform activity and answer rates.
topLevelTag	First tag of the question indicating the main topic and community that sees the question
numTags	Number of tags on the question. Treated and control questions are matched on the exact tag count, so they pose comparably scoped problems.
<i>Propensity score matching</i>	
userTenure	Number of days since the user’s first recorded activity on the platform, capturing cumulative experience.
numQuestionsAskedAT	Total number of questions asked by the user prior to the observation window.
numHelpProvidedAT	Total number of answers provided by the user prior to the observation window.
numQuestionsAsked30D	Number of questions asked in the 30 days preceding the observation window.
numHelpProvided30D	Number of answers provided in the 30 days preceding the observation window.
numQuestionsAsked7D	Number of questions asked in the 7 days preceding the observation window.
numHelpProvided7D	Number of answers provided in the 7 days preceding the observation window.
tagAcceptShareAvg	Average share of questions that receive an <i>accepted</i> answer within the question’s tags, capturing topic-specific difficulty.
<i>Observable question and author characteristics (propensity score matching)</i>	
postHour	Hour of day at which the question was posted, capturing diurnal variation in who is online to answer.
postDayOfWeek	Day of week on which the question was posted.
bodyLenChars	Length of the question body in characters, a proxy for question detail and effort.
titleLenChars	Length of the question title in characters.
ownerReputation	Asker’s reputation score at posting, capturing standing and experience beyond raw tenure.

Table 3: Covariate Balance Before and After Matching

Covariate	Unmatched			Matched		
	Tr Mean	Ct Mean	SMD	Tr Mean	Ct Mean	SMD
userTenure	570.95	720.62	−0.167	568.37	544.76	0.026
numQuestionsAskedAT	31.11	26.00	0.059	30.34	28.04	0.027
numHelpProvidedAT	15.27	14.15	0.009	14.36	12.15	0.018
numQuestionsAsked30D	2.00	1.19	0.197	1.96	1.84	0.030
numHelpProvided30D	0.81	0.46	0.068	0.76	0.62	0.028
numQuestionsAsked7D	0.57	0.34	0.174	0.56	0.53	0.023
numHelpProvided7D	0.22	0.12	0.063	0.20	0.16	0.026
tagAcceptShareAvg	0.51	0.44	0.695	0.51	0.51	−0.003
numTags	2.83	3.04	−0.173	2.83	2.83	0.000
bodyLenChars	1289.10	1658.54	−0.183	1286.37	1338.86	−0.026
titleLenChars	53.80	57.50	−0.172	53.79	53.97	−0.009
ownerReputation	4300.57	2541.81	0.115	4130.42	3499.00	0.041

Note. SMD = standardized mean difference. The conventional threshold for acceptable balance is $|\text{SMD}| < 0.1$. All post-matching SMDs fall well below this threshold.

4.5. Analytical Strategy: Cox Proportional Hazards with Time-Varying Covariates

We model the hazard of helping using a Cox proportional hazards model with time-varying covariates. The counting-process formulation of Andersen and Gill (1982) extends the original Cox model (Cox, 1972) to settings with time-varying covariates and recurrent events, providing the large-sample theoretical foundation (including consistency and asymptotic normality of the partial likelihood estimator) that justifies our approach.

For each matched pair of observations (one treated, one control), we partition the 4-day observation window into intervals defined by the key event times: the start of the window, the posting of the question, and the arrival of the answer (the actual answer arrival for treated users or the synthetic transition for controls).

The hazard function is specified as:

$$h(t \mid \mathbf{X}(t)) = h_0(t) \exp \left(\beta_0 \cdot \text{treatment} + \beta_1 \cdot \text{phasePostQuestion}(t) + \beta_2 \cdot \text{treatedPostQuestion}(t) \right. \\ \left. + \beta_3 \cdot \text{phasePostAnswer}(t) + \beta_4 \cdot \text{isTreatedActive}(t) \right) \quad (1)$$

where $h_0(t)$ is the baseline hazard. *treatment* is a time-constant indicator for treatment group membership (1 if the user’s question received an answer, 0 otherwise); it absorbs stable between-group differences in baseline helping rates. The remaining covariates are time-varying indicators that switch on at the appropriate event times. Because *treatedPostQuestion* switches on when the question is posted and remains on through the post-answer phase, the treated–control gap in the helping hazard equals β_2 during the waiting period and $\beta_2 + \beta_4$ after the answer arrives. The coefficient β_4 is therefore the difference-in-differences estimate of the change at answer arrival: the shift in the treated–control gap at the moment the answer lands, net of whatever divergence preceded it. We report β_4 as the treatment effect. The waiting-period coefficient β_2 is reported separately as a parallel-trends diagnostic: because no answer has yet arrived during the waiting period, it cannot reflect a response to answer receipt — it measures anticipatory or selection-driven divergence, carries no sign guarantee, and is accordingly not folded into the effect estimate.

To test the moderation by response time (Hypothesis 3), we extend the model with two additional terms:

$$h(t \mid \mathbf{X}(t)) = h_0(t) \exp \left(\boldsymbol{\beta}' \mathbf{X}(t) + \gamma \cdot \text{isTreatedActive}(t) \times \widetilde{\log}(\text{RT}) + \delta \cdot \text{treatedPostQuestion}(t) \times \widetilde{\log}(\text{RT}) \right) \quad (2)$$

where $\widetilde{\log}(\text{RT})$ denotes the standardized log response-time variable (see below), γ captures the moderation of the answer-arrival increment by answer speed, and δ captures the analogous moderation during the waiting period. Just as the base treatment effect is the sum $\beta_2 + \beta_4$, the response-time moderation of that effect is the sum $\gamma + \delta$; we report this net moderation alongside its two components. A negative $\gamma + \delta$ would indicate that faster answers strengthen the reciprocity effect (consistent with H3), while a positive sum would indicate the opposite.

Stratification by Tenure.. To test Hypothesis 2, we fit separate models for each of the seven tenure buckets. This allows the baseline hazard and all coefficients to vary freely across levels of user experience, providing a non-parametric view of how reciprocity changes over the user trajectory.

Identification. Our identification strategy rests on the matched DiD logic embedded in the model. Propensity score matching (Rosenbaum and Rubin, 1983) ensures that treated and control observations come from users with comparable baseline characteristics, including tenure, activity levels, and question topic difficulty. This matching absorbs between-user differences that might otherwise confound the treatment effect. We do not include user fixed effects in the survival models. Because matching is performed across different users (not within-user), and because the Cox model already leverages within-observation temporal variation through the time-varying covariate structure (Andersen and Gill, 1982), user fixed effects are not required for identification. The DiD structure compares within-observation changes before and after the answer across treatment groups, isolating the treatment effect from stable individual differences. The waiting-period covariate (*treatedPostQuestion*) captures any lift in helping that treated users show after asking but before receiving an answer. When this term is near zero, treated and control users move in parallel up to the answer, supporting the parallel-trends logic under which the arrival increment β_4 can be read as the effect of answer receipt rather than of the act of asking itself; when it is not, the two groups diverge before treatment arrives, and although β_4 nets out the average waiting-period gap, we flag such strata and interpret their estimates with added caution rather than treating the divergence as part of the effect. Even a near-zero waiting-period term, however, rules out only motivational shifts that materialize *before* the answer: question-induced channels that operate mainly after the answer arrives — most notably the elevated profile visibility a question generates — are not absorbed by it (see Section 6.5). We therefore report the waiting-period term explicitly, so that departures from parallel trends are measured rather than assumed away, and we return to its magnitude in the Results.

5. Results

5.1. Descriptive Statistics

The Cox analytic sample spans 24,349,760 question-level observations from 3,732,706 unique users on Stack Overflow, with 50% of questions receiving at least one answer. Help-event counts (answers the focal user posts to other users’ questions within the window) are right-skewed: the mean is [[AO:desc-mean-events]] events per observation window (SD = [[AO:desc-sd-events]]; median 0), consistent with helping being a relatively rare behavior

for any individual user in any given window. Among treated users, the median response time from question to first answer is 0.34 hours, though the distribution is heavily right-skewed (mean 5.08 hours, SD = 17.06), reflecting a minority of questions that wait many hours or days for a response. The sample spans all levels of platform experience, with the largest stratum being established users (1–3 years: $n = 6,424,968$), followed closely by the 1–6-month ($n = 4,045,952$) and < 1-week ($n = 4,029,008$) groups.

Table 4: Descriptive Statistics

Variable	Mean	SD	Median	N
Questions (observations)				24,349,760
Unique users				3,732,706
% questions with answer	50.00%			
Help events per window (answers + comments)	0.31	2.20	0.00	
User tenure (days)	656	839	328	
Response time (hours, treated)	5.08	17.06	0.34	
<i>Response time (hours, treated) by Tenure Bucket</i>				
< 1 Week	4.51	15.73	0.32	2,014,406
1 Week - 1 Month	3.59	13.83	0.26	775,751
1 - 6 Months	3.83	14.45	0.26	2,022,884
6 - 12 Months	4.24	15.42	0.28	1,491,553
1 - 3 Years	5.11	17.17	0.34	3,212,434
3 - 6 Years	6.69	19.87	0.50	1,807,077
> 6 Years	8.66	22.72	0.75	850,441
<i>N (unique questions) by Tenure Bucket</i>				
< 1 Week				4,029,008
1 Week - 1 Month				1,551,586
1 - 6 Months				4,045,952
6 - 12 Months				2,983,180
1 - 3 Years				6,424,968
3 - 6 Years				3,614,184
> 6 Years				1,700,882

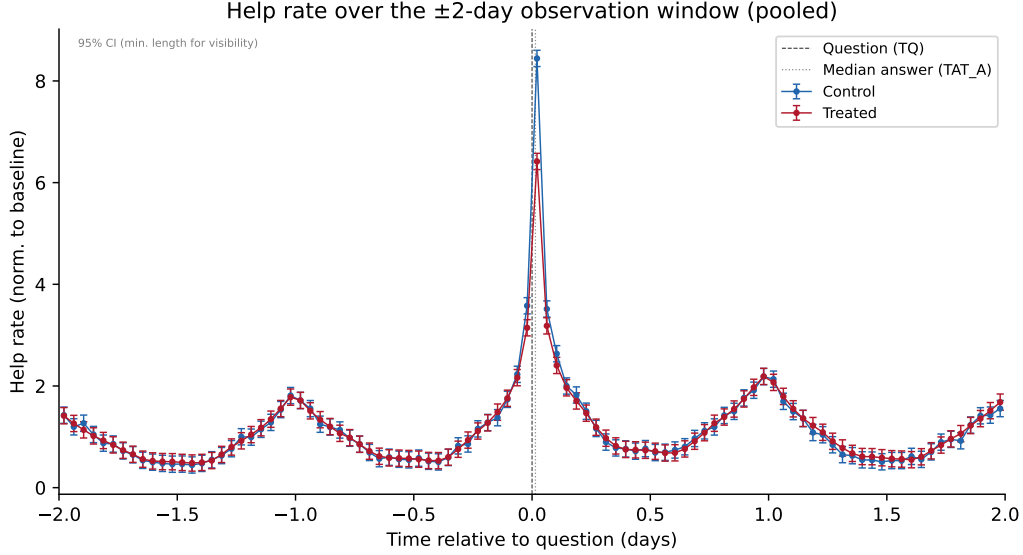


Figure 2: Help rate over the ± 4 -day observation window, pooled across all tenure buckets. The normalized help rate (relative to each group’s pre-question baseline) is plotted for treated users (red, who received an answer) and matched control users (blue, who did not).

5.2. Descriptive Evidence: Help Rates and the Parallel Trends Assumption

Figure 2 plots the normalized help rate for treated and control users over the two days before and after question posting. Before the question ($t < T_Q$), treated and control help rates track each other closely, consistent with the parallel-trends assumption and with the effectiveness of propensity score matching. At the moment of posting, both groups’ help rates spike sharply — users who have just asked a question are, mechanically, online and active — and the spike is more pronounced for controls, who continue waiting for an answer, than for treated users, whose questions are typically answered within the hour. Beyond this mechanical spike, the treated–control differences implied by our estimates (a few percent of the helping hazard) are too small to be visible in raw normalized rates, which is why the model-based difference-in-differences estimates below are needed to resolve them. We return to the control group’s larger at-question spike when discussing the response-time pattern.

5.3. Generalized Reciprocity Effect (H1)

To test Hypothesis 1, we estimate the pooled Cox model across all experience levels (Table 5). The treatment effect is the difference-in-differences increment at answer arrival (β_4); the waiting-period term (β_2) is reported alongside it as a parallel-trends diagnostic. The arrival increment is positive, statistically significant, and small: HR = 1.03 (95% CI [1.02, 1.05], $p < .001$), an approximately 3% increase in the instantaneous rate of helping after the answer arrives. The waiting-period diagnostic is essentially zero ($\beta_2 = \text{[[AO:pooled-beta2]]}$, n.s.), indicating that treated and control users moved in parallel until the answer arrived. Matched-pair bootstrap uncertainty for the pooled model, which preserves the matched structure, is reported in Appendix Table D.13 and `[[AO:boot-consistency — confirms/qualifies]]` the model-based interval. Hypothesis 1 therefore receives support in a deliberately modest form: the effect is far smaller than prior observational reports, and, as we show below, it is sensitive to conditioning on the quality of the answer received (Table 10).

A waiting-period term this close to zero indicates that the act of asking, by itself, did not shift treated users’ helping during the wait; it does not rule out question-induced influences — such as the increased profile visibility that a question generates — that persist into the post-answer window (see Section 6.5). As we show next, the modest pooled effect masks substantial heterogeneity: the arrival increment is largest for newcomers and vanishes with tenure. We turn first to the decomposition of the outcome by help type.

Table 5: Pooled Cox Regressions (All Experience Levels)

	Main	Main + Speed
<i>Treatment effect: increment at answer arrival (β_4)</i>		
Answer arrival ($\times$ Post-Answer Received)	0.0024 (0.0067)	0.0269*** (0.0058)
Hazard Ratio [95% CI]	[0.99, 1.02]	[1.02, 1.04]
<i>Total elevation (summed DiD, upper bound)</i>		
Received Answer (net post-answer effect)	0.0081† (0.0048)	0.0239*** (0.0045)
Hazard Ratio [95% CI]	[1.00, 1.02]	[1.02, 1.03]
Hazard Ratio [bootstrap 95% CI]	[1.01, 1.03]	—
<i>Pre-trend diagnostic (nested time-varying terms)</i>		
Waiting period (β_2): Received Answer $\times$ Post-Question	0.0056	-0.0030
<i>Response Time Interaction</i>		
Net: Treatment $\times$ log(RT), summed	—	0.0014 (0.0036)
Component: Post-Answer $\times$ log(RT)	—	-0.0780*** (0.0038)
N	24,349,760	24,349,760
Events	5,025,482	5,025,482

Cox-table standard errors are model-based; matched-pair bootstrap 95% CIs for the pooled DiD are in Table D.13.

Events = helping events in the full analysis sample (after time rounding). When interval rows exceed the fit cap, estimation uses a matched-pair subsample, but Events still refer to the full sample.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.1$

In response to the concern that an answers-only outcome may understate reciprocity by excluding lower-cost helping acts, Table 6 re-estimates the pooled model under broader outcome definitions. The decomposition confirms that the reciprocity signal is concentrated in *answers*: comments alone show no elevation (arrival HR = 1.00, 95% CI [0.98, 1.01]), and adding comments to answers dilutes rather than amplifies the estimate (answers-plus-comments arrival HR $\approx$ 1.00, [0.99, 1.02]) — the opposite of what a mechanical undercount of low-cost reciprocity among novices would produce. Edits of other users' posts, reported separately because they index curation more than costly contribution, carry a larger association (edits-only arrival HR = 1.14), and the all-channel composite sits between (answers-plus-comments-plus-edits arrival HR = 1.02). The waiting-period diagnostic is

near zero in every column except edits ($\beta_2 = +0.07$), and the effect enters at answer arrival, mirroring the pooled pattern. The answers-only estimate we headline is thus not an artifact of ignoring low-cost channels: the channel closest to conversational helping (comments) contributes nothing, and the one channel that would raise the estimate (edits) indexes a different construct.

Table 6: Reciprocity Effect Decomposed by Help Type

Outcome	Arrival HR [95% CI] (β_4)	Summed HR (upper bnd)	Waiting β_2	Events
Answers only	1.03 [1.02, 1.05]	1.03 [1.02, 1.04]	-0.00	3,412
Comments only	1.00 [0.98, 1.01]	1.01 [1.00, 1.03]	+0.02	2,109
Edits only	1.14 [1.12, 1.16]	1.23 [1.21, 1.25]	+0.07	987
Answers + comments	1.00 [0.99, 1.02]	1.01 [1.00, 1.02]	+0.01	5,012
Answers + comments + edits	1.02 [1.01, 1.03]	1.03 [1.02, 1.04]	+0.01	5,601

Arrival HR is the answer-arrival increment $\exp(\beta_4)$ (primary); Summed HR is the DiD $\exp(\beta_2 + \beta_4)$ (upper bound); β_2 is the waiting-period (anticipatory-engagement) pre-trend. Accepting an answer is excluded from the decomposition: it is a self-directed act available only to treated users (a control never receives an answer to accept), so the matched difference-in-differences contrast is degenerate for it. Cox-table standard errors are model-based; matched-pair bootstrap uncertainty is reported separately when generated. Events = helping events in the full analysis sample (after time rounding). When interval rows exceed the fit cap, estimation uses a matched-pair subsample, but Events still refer to the full sample.

5.4. Experience Moderation of Reciprocity (H2)

To test Hypothesis 2, we estimate the model separately within each of the seven tenure buckets (Table 7; Figure 3). The arrival increment is steeply graded by tenure, weakening from newcomers to veterans. It is largest for the newest users — in the first week, receiving an answer raises the post-answer helping hazard by about $[[AO:nc-arrival-pct]]\%$ (arrival HR = $[[AO:nc-arrival-hr]]$, 95% CI $[[AO:nc-arrival-ci]]$, $p < .001$) — remains positive and statistically significant through the intermediate strata (arrival hazard ratios of roughly 1.03–1.05) and, more weakly, at three to six years (HR = $[[AO:b6-arrival-hr-ci-sig]]$), and reaches zero only among users with more than six years of tenure (HR = $[[AO:b7-arrival-hr-ci-sig]]$, n.s.). This graded pattern offers clear support for Hypothesis 2: the reciprocity-like elevation is concentrated at the start of the user trajectory and fades as experience accumulates.

For newcomers, `[[AO:nc-beta2-beta4-pattern — verify: are both the arrival increment ( $\beta - 4$ ) and the waiting-period diagnostic ( $\beta - 2$ ) positive and significant?]]`, whereas among veterans both terms are near zero. `[[AO:if nc-beta2 positive, keep: A positive waiting-period term means part of newcomers' elevation begins before the answer arrives; the arrival increment is measured net of that divergence, but a nonzero pre-trend in this stratum is itself a caution that anticipation, exposure, or selection channels are active — a pattern we return to when discussing the exposure and selection channels the design cannot fully separate from reciprocity (Section 6.5).]]`

The tenure gradient is not an artifact of pooling structurally different entry cohorts. Because calendar year enters the design only as an exact-matching stratum, treated and control questions within a pair always come from the same year; to probe heterogeneity *across* years we re-estimate Model A on the pre-2020 subsample, before the generative-AI disruptions of 2022–2023, both pooled and among newcomers. The pooled pre-2020 arrival hazard ratio is `[[AO:pre2020-pooled-hr]]` (versus 1.03 in the full sample), and the newcomer association is `[[AO:pre2020-nc-pattern — essentially unchanged]]` (arrival HR = `[[AO:pre2020-nc-hr]]` pre-2020 versus `[[AO:nc-arrival-hr]]` in the full sample; Appendix Table E.14), indicating that the tenure gradient was already present on the pre-generative-AI platform rather than driven by recent cohorts.

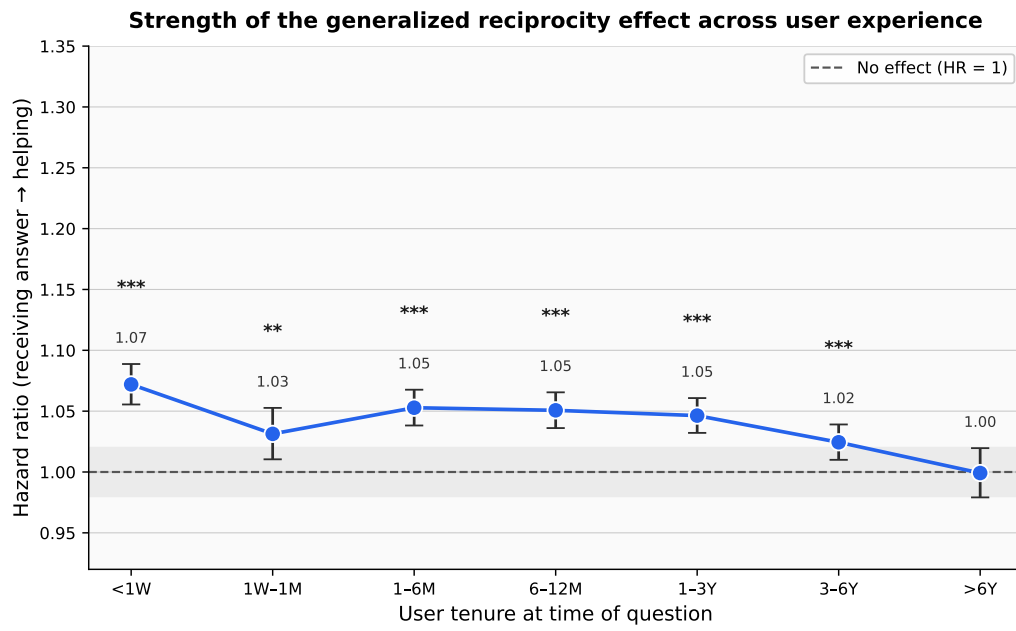


Figure 3: Strength of the generalized reciprocity effect across user experience. Bars show the hazard ratio for the treatment effect by tenure bucket. Error bars indicate 95% confidence intervals.

Table 7: Cox Regression Results: Effect of Receiving an Answer on Helping Hazard

	< 1 Week	1 Wk– 1 Mo	1–6 Mo	6–12 Mo	1–3 Yr	3–6 Yr	> 6 Yrs
<i>Treatment effect: increment at answer arrival (β_4)</i>							
Answer arrival ($\times$ Post-Answer Received)	0.0573***	0.0206*	0.0419***	0.0412***	0.0268***	0.0078	-0.0191*
	(0.0078)	(0.0100)	(0.0068)	(0.0066)	(0.0064)	(0.0065)	(0.0090)
<i>Hazard Ratio</i> [95% CI]	[1.04, 1.08]	[1.00, 1.04]	[1.03, 1.06]	[1.03, 1.06]	[1.01, 1.04]	[0.99, 1.02]	[0.96, 1.00]
<i>Total elevation (summed DiD, upper bound)</i>							
Received Answer (net post-answer)	0.1018***	-0.0121†	0.0215***	0.0258***	0.0272***	0.0032	-0.0066
	(0.0062)	(0.0068)	(0.0045)	(0.0044)	(0.0045)	(0.0050)	(0.0076)
<i>Hazard Ratio</i> [95% CI]	[1.09, 1.12]	[0.97, 1.00]	[1.01, 1.03]	[1.02, 1.04]	[1.02, 1.04]	[0.99, 1.01]	[0.98, 1.01]
<i>Pre-trend diagnostic (nested time-varying terms)</i>							
Waiting period ($\beta_2, \times$ Post-Question)	0.0445***	-0.0327**	-0.0204**	-0.0154*	0.0004	-0.0046	0.0125
N	4,029,008	1,551,586	4,045,952	2,983,180	6,424,968	3,614,184	1,700,882
Events	475,633	310,464	947,070	760,087	1,571,611	717,699	242,918

N = unique questions (treated + control) in the Cox sample. Within each column, treated vs. control counts can differ because tenure is defined per question.

Cox-table standard errors are model-based; matched-pair bootstrap 95% CIs for the pooled DiD are in Table D.13.

Events = helping events in the full analysis sample (after time rounding). When interval rows exceed the fit cap, estimation uses a matched-pair subsample, but Events still refer to the full sample.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.1$

5.5. Response Time Moderation (H3)

Hypothesis 3 predicted that faster responses would strengthen the reciprocity effect. Table 8 reports the arrival increment and its response-time moderation, estimated separately by tenure bucket, and Table 9 and Figure 4 give a more flexible discrete-bin view. The evidence offers qualified support for the immediacy prediction. The continuous log-response-time interaction with the arrival increment (γ) is negative in every tenure bucket and statistically significant in all but the most senior (about -0.07 to -0.09 across the newcomer through one-year buckets, -0.06 at one to three years, weakening to -0.02 at three to six years and to zero, n.s., among users with more than six years), meaning that a faster answer is associated with a larger increment at arrival — the direction H3 predicts. The discrete bins tell a consistent, if noisier, story: the arrival increment is near-null for answers within thirty minutes, peaks sharply at `[[AO:bin-3060-hr-ci]]` for answers arriving in the 30–60-minute window, and falls back to modest values from one hour to one day (roughly 0.98–1.05), with a smaller secondary elevation at multi-day delays (1–3 days: `[[AO:bin-1-3d-hr-sig]]`). The pronounced early peak is closer to the “sweet spot” pattern originally hypothesized than to either a monotone speed gradient or a flat null; we return to the late elevation below.

Two qualifications temper this reading. First, the waiting-period term carries a response-time moderation of its own (δ , `[[AO:delta-sign]]`) that `[[AO:delta-offset-pattern — verify: largely offsets the arrival moderation]]`, so the immediacy pattern is a property of the increment at answer arrival specifically, not of treated users’ overall activity trajectories. Second, we re-estimated each response-time bin with the same answer-quality controls (Appendix Table E.17). `[[AO:bin-3060-quality-verdict — verify: The 30–60-minute peak attenuates but remains the clear early-window maximum (X.XX to X.XX)]]`, so it is not merely an artifact of fast answers being better answers. The smaller multi-day elevation visible in the baseline bins `[[AO:late-bump-quality — persists/attenuates]]` under the same controls; because the quality controls are post-treatment — defined only for answered questions and entered as linear terms — we cannot cleanly interpret this late bump and do not build on it. The central point is that the early-window advantage survives adjustment for answer quality, so we read the surviving 30–60-minute elevation as a conservative lower bound and treat the immediacy pattern as supported, with the usual caution attaching to observational, quality-adjusted comparisons.

Table 8: Response Time Moderation of the Reciprocity Effect

	< 1 Week	1 Wk– 1 Mo	1–6 Mo	6–12 Mo	1–3 Yr	3–6 Yr	> 6 Yrs
<i>Base treatment effect: arrival increment (β_4, at mean response time)</i>							
Answer arrival (net post-answer)	-0.0128*	0.0083	0.0320***	0.0392***	0.0456***	0.0359***	0.0245**
	(0.0063)	(0.0083)	(0.0057)	(0.0057)	(0.0056)	(0.0058)	(0.0079)
<i>Base treatment effect (summed DiD, upper bound)</i>							
Received Answer, summed	-0.0239***	-0.0111†	0.0278***	0.0424***	0.0523***	0.0344***	0.0282***
	(0.0056)	(0.0064)	(0.0043)	(0.0042)	(0.0042)	(0.0046)	(0.0070)
<i>Response-time moderation</i>							
Answer arrival $\times$ log(RT) (γ)	-0.0742***	-0.0838***	-0.1054***	-0.1006***	-0.0757***	-0.0341***	-0.0216***
	(0.0053)	(0.0064)	(0.0041)	(0.0038)	(0.0035)	(0.0034)	(0.0046)
Net: Treatment $\times$ log(RT), summed ($\gamma + \delta$, upper bound)	-0.0521***	0.0018	0.0002	-0.0028	0.0102**	0.0246***	0.0206***
	(0.0054)	(0.0061)	(0.0039)	(0.0036)	(0.0033)	(0.0033)	(0.0046)
Waiting period $\times$ log(RT) (δ)	0.0221***	0.0856***	0.1055***	0.0978***	0.0859***	0.0587***	0.0423***
N	4,029,008	1,551,586	4,045,952	2,983,180	6,424,968	3,614,184	1,700,882
Events	475,633	310,464	947,070	760,087	1,571,611	717,699	242,918

Cox-table standard errors are model-based; matched-pair bootstrap 95% CIs for the pooled DiD are in Table D.13.
Events = helping events in the full analysis sample (after time rounding). When interval rows exceed the fit cap, estimation uses a matched-pair subsample, but Events still refer to the full sample.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.1$

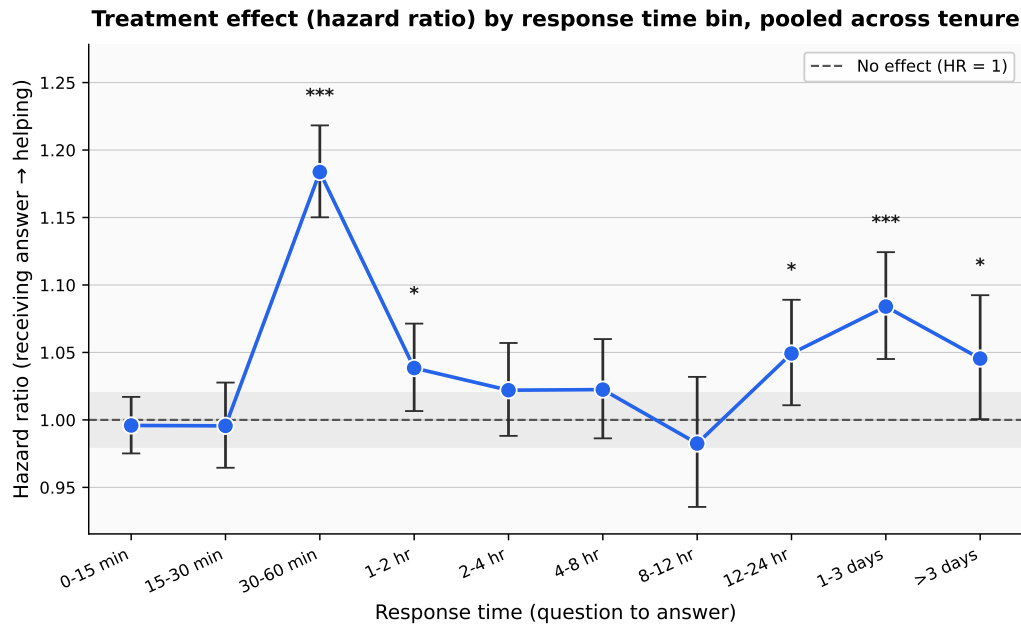


Figure 4: Treatment effect (hazard ratio) by response-time bin, pooled across all tenure buckets. Each point represents the estimated hazard ratio of the treatment indicator within that response-time bin, relative to the matched control. Error bars indicate 95% confidence intervals.

Table 9: Treatment Effect by Response-Time Bin

Response time	HR	95% CI	N	Events
0-15 min	0.99	[0.97, 1.01]	17,412,740	3,598,284
15-30 min	0.99	[0.96, 1.02]	13,934,456	2,655,887
30-60 min	1.22***	[1.19, 1.25]	13,549,736	2,559,608
1-2 hr	1.03*	[1.00, 1.06]	13,211,144	2,491,230
2-4 hr	1.03	[0.99, 1.06]	12,958,927	2,439,543
4-8 hr	1.00	[0.97, 1.03]	12,762,008	2,403,607
8-12 hr	0.97	[0.93, 1.01]	12,464,162	2,346,654
12-24 hr	1.04*	[1.00, 1.07]	12,645,932	2,384,160
1-3 days	1.04**	[1.01, 1.08]	12,592,667	2,400,019
>3 days	1.01	[0.98, 1.05]	12,391,574	2,390,995

Cox-table standard errors are model-based; matched-pair bootstrap 95% CIs for the pooled DiD are in Table D.13.

Events = helping events in the full analysis sample (after time rounding). When interval rows exceed the fit cap, estimation uses a matched-pair subsample, but Events still refer to the full sample.

Each row fits Model A to treated questions in that response-time bin plus the full no-answer control pool.

HR is the answer-arrival increment (β_4 , is_treated_active) at answer arrival; the summed DiD contrast is reported in the pooled regression tables.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.1$

Answer quality as an alternative explanation.. A natural concern, raised in review, is that response time proxies for answer quality: very fast answers are disproportionately terse or link-only, whereas answers that take longer to arrive tend to be fuller and more detailed, so it may be the quality of the answer — not its timing — that elicits reciprocal helping. We probe this directly by re-estimating the pooled speed specification with controls for three observable quality signals, re-extracted from the raw data dump for this revision: the length of the question’s first answer in characters, its vote score, and whether it was subsequently accepted (Table 10). Entered one at a time, they tell a graded story. Controlling for *answer length* alone — the reviewer’s terse-versus-detailed variable, and the signal least entangled with the asker’s own behavior, since it is fixed by the answerer at composition — [[AO:quality-length-effect — verify: barely moves the estimate]] (arrival increment [[AO:quality-baseline-hr]] without controls, [[AO:quality-length-hr]] with length only). Controlling for the answer’s vote score alone brings it to [[AO:quality-score-hr]], and the full specification, which additionally conditions on whether the answer was accepted, [[AO:quality-full-direction —

attenuates it further / pushes it below one]] (arrival [[AO:quality-full-hr]]), with the same pattern in the newcomer stratum (quality-controlled arrival [[AO:nc-quality-hr]]; Appendix Table E.16). The same three controls, applied within each discrete response-time bin rather than pooled, are reported in Appendix Table E.17 (discussed with the response-time results above). We read the fuller specification as a bounding exercise rather than a clean causal decomposition, because vote score and, especially, acceptance are post-treatment and endogenous to the outcome — accepting requires the asker to return to the site, the same re-engagement that produces the helping outcome, so conditioning on it partials out part of the very channel through which reciprocity would operate. The upshot is two-sided. The association is robust to the reviewer’s stated concern — answer length — and is reversed only by adjustment for the more endogenous signals, exactly as a bad-control logic anticipates; yet we cannot rule out that answer quality, imperfectly measured, is a genuine rival to any timing- or gratitude-based reading of the response-time pattern. The residual caveat is that character count is a coarse proxy for length: it does not distinguish worked code and explanation from boilerplate.

Table 10: Answer-Quality Robustness

Specification	Arrival HR	95% CI	Summed HR (upper bnd)	N	Events
speed_baseline	1.027	[1.016, 1.039]	1.024 [1.02, 1.03]	24,349,760	5,025,482
speed_length_only	1.022	[1.010, 1.033]	1.010 [1.00, 1.02]	24,349,760	5,025,482
speed_score_only	1.004	[0.993, 1.016]	0.978 [0.97, 0.99]	24,349,760	5,025,482
speed_quality_controls	0.982	[0.970, 0.993]	0.930 [0.92, 0.94]	24,349,760	5,025,482

Primary HR is the answer-arrival increment (β_4); the summed DiD ($\exp(\beta_2 + \beta_4)$) is shown as an upper bound.

Cox-table standard errors are model-based; matched-pair bootstrap uncertainty is reported separately when generated.

Events = helping events in the full analysis sample (after time rounding). When interval rows exceed the fit cap, estimation uses a matched-pair subsample, but Events still refer to the full sample.

One interpretation of the response-time pattern draws on platform session dynamics: users who post a question may remain actively engaged while waiting, elevating helping during the waiting period regardless of whether an answer arrives. On this account, a fast answer might discharge that pending engagement rather than trigger graded reciprocal gratitude. This

interpretation is speculative — we do not observe session boundaries (logins, logouts, inactivity gaps). The answer-quality account above is an equally plausible reading of the same pattern, and a third, purely selective account is that users whose questions attract answers were simply more active to begin with. The design cannot adjudicate among these, and we present the session-dynamics reading as one candidate among several rather than as an established mechanism.

Figure 5 provides descriptive context, comparing users who have received an answer (red) with users who will receive one but are still waiting (blue), alongside matched controls. In the first half-hour after the question — when only the fastest answers have arrived — users still waiting help at the highest rate, consistent with the mechanical elevation of users who have just asked a question and remain on the platform. From roughly an hour onward the ordering flips: users who have received an answer help at a modestly higher rate than those still waiting and than controls, consistent with a small post-answer elevation. Figure 6 disaggregates this comparison by tenure bucket and shows the same qualitative pattern in every stratum. These raw-rate orderings are descriptive: the differences are small, and answer-arrival timing is not random within the treated group, so we do not read them as independent evidence for any one mechanism — but they are consistent with the modest arrival increments the models estimate.

6. Discussion

6.1. Methodological Contribution

A persistent difficulty in studying generalized reciprocity from platform data is that the same user characteristics that predict seeking help also predict giving it. Active, engaged users are more likely both to post questions and to post answers, so correlations between help received and help given can reflect differences in user engagement rather than any causal response to receiving help. Prior observational studies that report positive associations between help-seeking and subsequent helping behavior without accounting for this selection (Baker and Bulkley, 2014; Chen et al., 2018; Joyce and Kraut, 2006) may have overstated the role of reciprocity. The matched DiD survival design developed here addresses this confound by comparing changes in helping behavior, before and after question posting, between users whose questions received answers and propensity-score-matched users whose questions did not. Entering the waiting period as an explicit term makes any

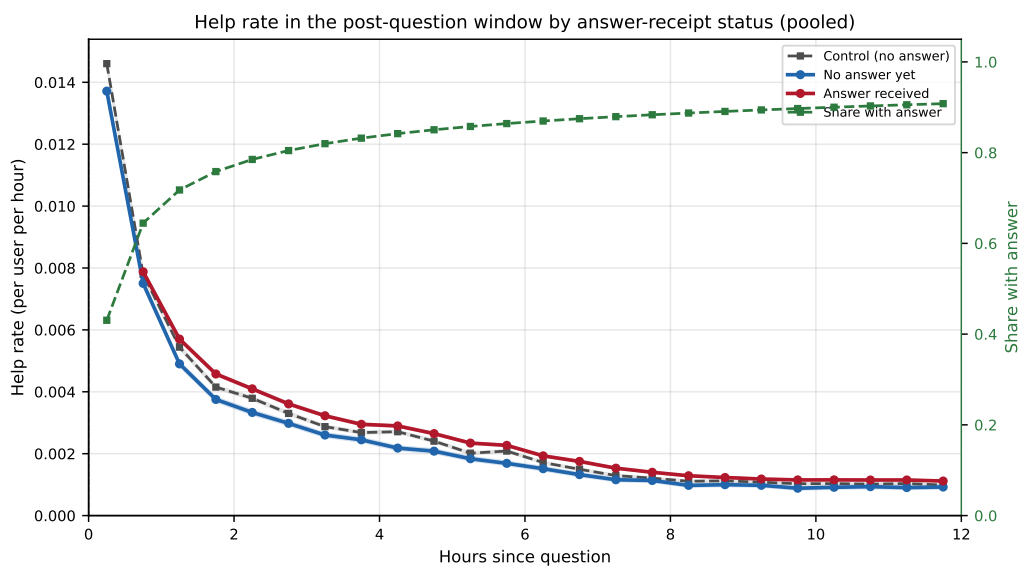


Figure 5: Help rate in the post-question window by answer-receipt status, pooled across all tenure buckets. Grey dashed: matched control users who received no answer. Blue: treated users who have not yet received an answer (No answer yet). Red: treated users who have already received an answer (Answer received). Green dashed (right axis): cumulative share of treated users who have received an answer by each hour.

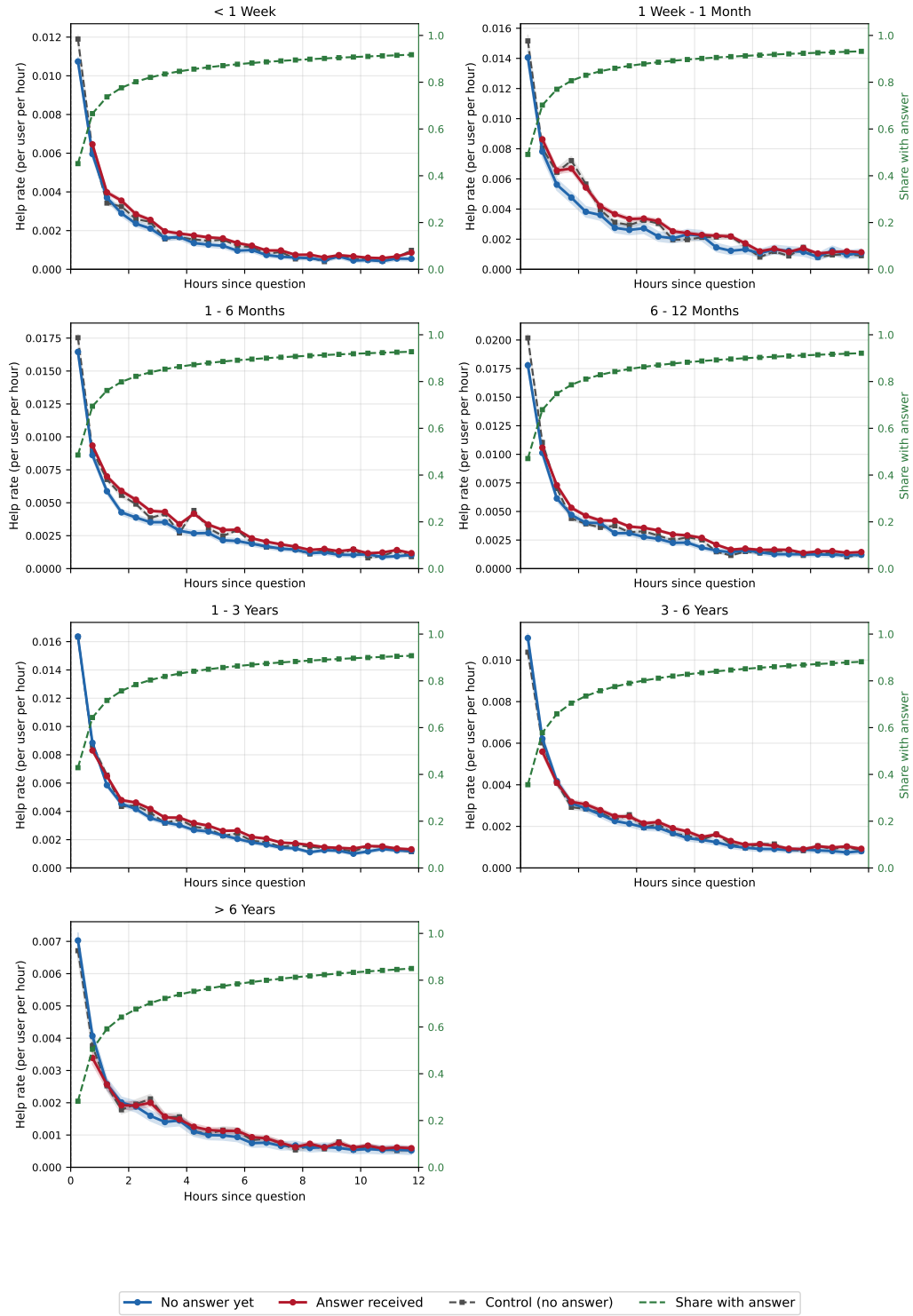


Figure 6: Help rate in the post-question window by answer-receipt status, separately by tenure bucket. Grey dashed: matched control users who received no answer. Blue: treated users who have not yet received an answer (No answer yet). Red: treated users who have already received an answer (Answer received). Green dashed (right axis): cumulative share of treated users who have received an answer by each hour.

pre-answer divergence visible rather than silently absorbed into the treatment effect. In the present specification, the pooled contrast is small (arrival-increment $HR = 1.03$) and the association steeply newcomer-concentrated — substantially smaller than many earlier reports, and, as the answer-quality check shows, sensitive to controlling for the quality of the answer received, consistent with substantial confounding in prior designs.

6.2. Reciprocity as a Contributor-Recruitment Mechanism

The treatment effect shows a clear tenure gradient: the arrival increment is largest among the newest users ($HR = [[AO:nc-arrival-hr]]$) and falls to null among the most experienced (arrival $HR = [[AO:b7-arrival-hr]]$). We cannot directly observe whether this gradient reflects reciprocity, selection into being answered, the quality of the answers received, or the displacement of general moral motives by community-specific incentives — the design identifies behavioral associations, not motivational shifts. If the gradient does reflect reciprocity rather than these alternatives, it would imply a narrow role for generalized reciprocity: an early association concentrated at the point of entry, before community-specific motivations are established.

As users accumulate experience, community-specific incentives may coincide with — and plausibly displace — the general reciprocal impulse, though we cannot directly observe this motivational shift. This is consistent with theories of norm displacement (Bénabou and Tirole, 2006), which show how the introduction of external incentives can crowd out moral motives. On Stack Overflow, reputation systems and status markers provide salient, community-specific reasons to help that are not available to newcomers. The null association among the most experienced users is consistent with this account, though it is equally consistent with selection into being answered in that stratum.

This reframing also helps account for the inconsistency of findings in the prior literature. Studies that draw primarily on samples of experienced users, or that aggregate across experience levels dominated by experienced contributors, should find weaker reciprocity effects under this account. Studies that capture behavior early in the user trajectory should find stronger ones. The variation in prior findings may thus reflect not noise or methodological failure, but genuine heterogeneity in the effect across the user population.

6.3. *The Response-Time Pattern and Its Interpretation*

The response-time analysis was intended to test the prediction, derived from laboratory studies of gratitude, that faster help produces stronger reciprocity. The pattern offers it qualified support. The continuous interaction runs in the predicted direction — the arrival increment is larger for faster answers in every tenure bucket — and the discrete bins locate the effect in an early response-time window, with a pronounced peak for answers arriving within thirty to sixty minutes. This is broadly consistent with an emotional-immediacy account, in which help that arrives while the need is still salient elicits a stronger reciprocal response. Two features counsel caution. The very fastest answers (within thirty minutes) show no elevation, which a pure immediacy account does not predict; and the same specification’s base effect `[[AO:quality-full-direction-discussion — attenuates markedly / falls below one]]` once answer quality is controlled, so we cannot rule out that the timing pattern partly reflects the quality of fast versus slow answers rather than their speed per se.

Two non-gratitude readings could also generate an early-window peak. The first is answer quality: answers that arrive within the first few minutes may be terse or link-only, while those in the 30–60-minute window may be substantive enough to prompt reciprocation; controlling for observable quality does `[[AO:quality-channel-discussion — move the base effect toward/below one]]`, consistent with this channel. The second is platform session dynamics: an answer that arrives while a user is still in an active session may catch them at a moment of elevated engagement, whereas an answer that arrives after the session has ended cannot. We advance both only as conjectures; our data cannot cleanly distinguish them from a genuine immediacy effect or from selection, because we observe neither answer text nor the session boundaries (logins, logouts, inactivity gaps) that a direct test would require. Conditioning the response-time bins on answer quality (Appendix Table E.17) `[[AO:bin-3060-quality-discussion — verify: leaves the 30–60-minute peak intact — it attenuates but remains the clear early maximum (X.XX to X.XX) —]]` so the early-window advantage is not an artifact of fast answers being better answers. A weaker secondary elevation at multi-day delays also appears, but because it sits far from the immediacy window and the quality controls are post-treatment, we cannot attach a clean mechanism to it and do not build on it. The immediacy pattern is thus supported for the early window, subject to the usual caution about observational, quality-adjusted comparisons.

The response-time pattern bears only weakly on the norm-displacement account. The arrival increment and its response-time moderation are both near zero among the most experienced users, which is at least consistent with the mechanism as a whole having faded rather than merely dampened. We do not lean on the response-time pattern as evidence for displacement.

6.4. *Implications for Platform Design*

Because any reciprocity-like elevation is modest and concentrated among newcomers, the practical leverage platforms can derive from it lies primarily in contributor recruitment rather than in sustaining contributions from established members — and even that recruitment effect is small in absolute terms.

The most direct implication is that platforms seeking to grow their contributor base should prioritize ensuring that new users receive answers to their questions. In absolute terms, the newcomer arrival increment (HR = [[AO:nc-arrival-hr]]; Table 7) translates to an additional helping-event risk of roughly [[AO:nc-ard-pp]] percentage points per observation window, with a number-needed-to-treat of about [[AO:nc-nnt]]: roughly [[AO:nc-nnt]] newcomer questions must receive an answer to yield one additional helping event to others in the window (Table D.12). The pooled effect across all tenure levels is far smaller (number-needed-to-treat near [[AO:pooled-nnt]]) and sensitive to answer-quality controls, so the design implication is not a platform-wide helping multiplier but a targeted — and modest — newcomer onboarding channel.

Our results provide only tentative support for a response-time lever. The reciprocity signal is concentrated in answers that arrive early, with a peak in the 30–60-minute window (Tables 8 and 9), which is consistent with prioritizing prompt answers to newcomers. But because the effect is small and we cannot fully separate answer speed from answer quality — even though the 30–60-minute advantage does survive quality controls — we stop short of recommending interventions built specifically on response speed. The better-supported emphasis is answer receipt itself — ensuring that newcomers’ questions are answered at all, and promptly, and that answer notifications are salient — rather than optimizing for a particular response-time window.

A further implication concerns the increasing prevalence of AI-generated answers on question-and-answer platforms (see also the GAI-era limitation below). If AI assistance substitutes for human responses, the human-to-human interaction that could sustain generalized reciprocity may decline.

Given that any residual association is concentrated among newcomers, this substitution could weaken an early onboarding channel through which new users currently receive a modest push into helping from human community members.

6.5. Limitations

Several limitations constrain the conclusions that can be drawn from this analysis.

The most important identification caveat is right-truncation and churn: users who post a question and leave the platform before the fixed post-question window ends are under-observed for the outcome in ways correlated with both treatment and helping. We do not implement a Heckman-type correction; instead, Appendix Table D.11 reports transparent bounds on the contrast under alternative assumptions about zero-event controls. Even under optimistic bounds, the pooled effect remains modest; the newcomer contrast is more sensitive to the churn assumption and should be read with that sensitivity in mind.

Our helping outcome is the posting of an answer to another user’s question (self-directed accepts excluded); the decomposition (Table 6) shows this choice is not what generates the effect — comments alone carry no elevation, and adding them dilutes the estimate. Two further low-cost reciprocal channels that novices may favor nonetheless sit outside the outcome: edits of others’ posts, which we observe and report separately (edits-only arrival HR 1.14) but keep out of the headline outcome because they index curation more than costly contribution, and upvotes, which are absent from our processed data and would require re-parsing the raw dump to recover. To the extent that newcomers reciprocate disproportionately through these unmeasured or separately tabulated channels, our answers-anchored estimates do not capture that portion of reciprocal behavior, and the reported hazard ratios describe costly contribution specifically rather than a complete accounting of reciprocity. We also cannot cleanly separate a reciprocal response to receiving help from selection into being answered.

Beyond identification, the design captures a behavioral association but cannot identify the psychological mechanism producing it. Gratitude, moral obligation, incentive displacement, and platform-session dynamics are all *proposed* interpretations consistent with parts of the pattern; the data do not allow us to distinguish among them, and we do not observe session boundaries, reputation salience, or emotional states. A particularly consequential

competing explanation is answer quality. Faster answers are systematically terser than slower ones; while the pooled association is [[AO:quality-length-limitation — verify: robust to controlling for answer length alone]], adding the more endogenous quality signals — vote score and, especially, whether the first answer was accepted — [[AO:quality-full-limitation — attenuates the pooled base effect markedly / pushes it below one]] (Table 10), so we cannot attribute the association to answer receipt net of answer quality. These controls are imperfect: acceptance and score are post-treatment and partly endogenous (accepting requires the asker to return to the site), and character count captures answer length only coarsely, without distinguishing worked code and explanation from filler. The direction is nonetheless a genuine caveat, and it counts against a quality-free reading of the effect. Future work combining behavioral records with experience-sampling measures of emotional state or moral reasoning during platform use could help disentangle these pathways.

The identification strategy rests on the assumption that, conditional on the matched covariates and the DiD structure, answer receipt is approximately independent of the potential outcomes for helping behavior. The propensity score matching substantially improves the comparability of treated and control observations, as confirmed by post-matching balance diagnostics. Unobserved time-varying factors, including changes in workload, access to alternative information sources, or exogenous variation in a user’s availability, could nonetheless bias the estimates in ways that the design cannot rule out.

A specific unobserved channel of this kind is exposure. Posting a question raises the asker’s profile visibility: other users who encounter the question may view the asker’s profile and contribution history, and this salience can generate reputation concern or social-obligation pressure that is distinct from gratitude for a received answer yet persists into the post-answer window. The waiting-period covariate (*treatedPostQuestion*) absorbs the baseline effect of the act of asking, but not any residual visibility that continues after the answer arrives, so the design cannot fully separate gratitude for answer receipt from this exposure channel. The threat is to mechanism attribution rather than to the existence of the behavioral pattern: the arrival increment remains a well-defined estimate of the behavioral change at answer receipt, but part of any elevation could reflect exposure rather than reciprocity. A natural probe — a placebo test asking whether unanswered questions with *high* view counts show a helping lift in a pseudo-post-answer window — is unfortunately not

identified with the archival data available to us: the Stack Overflow data dump records *ViewCount* only as a single cumulative snapshot at the dump date, so a question’s view count conflates its age, topic popularity, and years of subsequent (largely search-engine) traffic with the contemporaneous exposure during the days-long observation window that such a placebo would require. Testing the exposure channel directly would require time-stamped view data or repeated dump snapshots that do not exist for our study period.

Response time may partially proxy for question characteristics not fully captured by the tag-level accepted-answer rate, since questions that attract fast responses may differ systematically from those that wait. Future work with random treatment or finer-grained measures of question difficulty, such as text-based complexity or specificity scores, could provide a cleaner test of the response-time mechanism.

The session-dynamics and re-engagement interpretations rest on indirect evidence: discrete-bin patterns, continuous speed interactions, and descriptive help-rate figures. Direct tests would require session-level data, including login and logout timestamps or inactivity-based session boundaries. We treat these accounts as speculative conjectures, not established mechanisms.

Stack Overflow is a large, relatively anonymous, gamified platform built around technical question-answering. The patterns we estimate may not transfer to smaller or relation-based communities (enterprise Slack channels, academic peer review, intimate hobby forums) where social ties and non-status motives differ. We therefore refrain from claiming that generalized reciprocity “likely operates” elsewhere; at most, our estimates describe one platform’s pre-/early-generative-AI era.

The analysis window spans Stack Overflow’s inception through 1 April 2025 and should be read as a *historical snapshot*, not a prescriptive guide for contemporary platform design. Since 2023, LLM-based tools have structurally disrupted Stack Overflow’s ecosystem: active question volume and human-to-human answering have declined sharply. The human-answer reciprocity associations we estimate were formed when community answering was the norm; their relevance to platforms increasingly mediated by AI-generated responses is uncertain. The pattern is not, however, confined to the recent AI-affected period: restricting the sample to the pre-2020 era leaves both the pooled association and the newcomer association essentially unchanged (Appendix Table E.14), so what we document characterizes the pre-generative-AI platform rather than a transient artifact of its disruption.

Finally, the analysis establishes that any short-run association declines

with experience but does not characterize what replaces it. The theoretical argument implicates community-specific norms and status motives, but testing this account would require integrating behavioral analysis with direct measures of those alternative mechanisms, for instance, through surveys or experimental manipulations of reputation visibility.

7. Conclusion

Generalized reciprocity is widely theorized as a mechanism sustaining cooperation on knowledge-sharing platforms, but observational evidence is limited by a persistent confound: active users are more likely both to seek help and to give it, regardless of whether their questions are resolved. Our matched difference-in-differences survival design confronts this confound directly. The pooled treatment effect is positive but small (difference-in-differences arrival increment $HR = 1.03$), steeply concentrated among newcomers (arrival $HR = [[AO:nc-arrival-hr]]$) and undetectable among veterans. The newcomer association is the most robust pattern in the data — it holds in the pre-2020, pre-generative-AI subsample as well as in the full sample (Appendix Table E.14) — yet even it is modest in absolute terms and $[[AO:quality-conclusion]]$ is sensitive to / does not survive controls for the quality of the answer received. Receiving help appears to nudge newcomers toward helping others, but our estimates cannot, on their own, separate that story from selection into being answered or from the quality of the help that arrives; distinguishing them remains a task for designs with exogenous variation in whether, and how quickly, help arrives.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Ethics statement

This study analyzed de-identified, publicly available archival records from Stack Overflow. No interaction with human participants occurred, and no personally identifiable information was collected or analyzed. Ethical approval was not required under applicable institutional policies for secondary analysis of public platform data.

Data availability

Stack Overflow public data dumps were obtained from the Internet Archive (<https://archive.org/details/stackexchange>). Analysis code and step-by-step instructions to reproduce all tables and figures are available in a public repository (URL withheld for blind review). Processed analytical datasets used in the matching and survival analyses are available from the corresponding author upon reasonable request.

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Appendix A. Replication materials

Analysis code and instructions to reproduce all tables and figures are available in a public repository (URL withheld for blind review).

Appendix B. Matching Diagnostics

Figure B.7 shows sufficient common support between the treatment and control groups: the propensity-score distributions of the original treated and control groups are similar, and the distribution of the matched control group is nearly identical to that of the treated group.

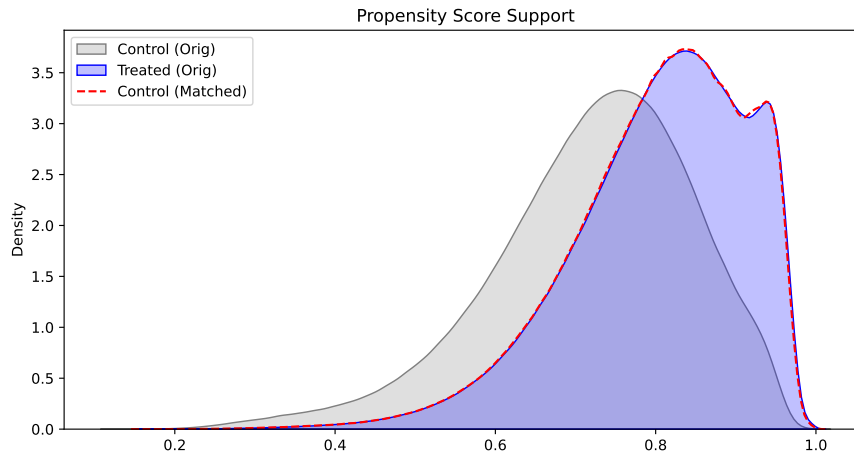


Figure B.7: Distribution of propensity scores of treatment group, control group, and matched control group

Figure B.8 displays the standardized mean differences (SMDs) for the matching covariates before and after matching. Several covariates show noticeable imbalance before matching; after matching, all SMDs are close to zero.

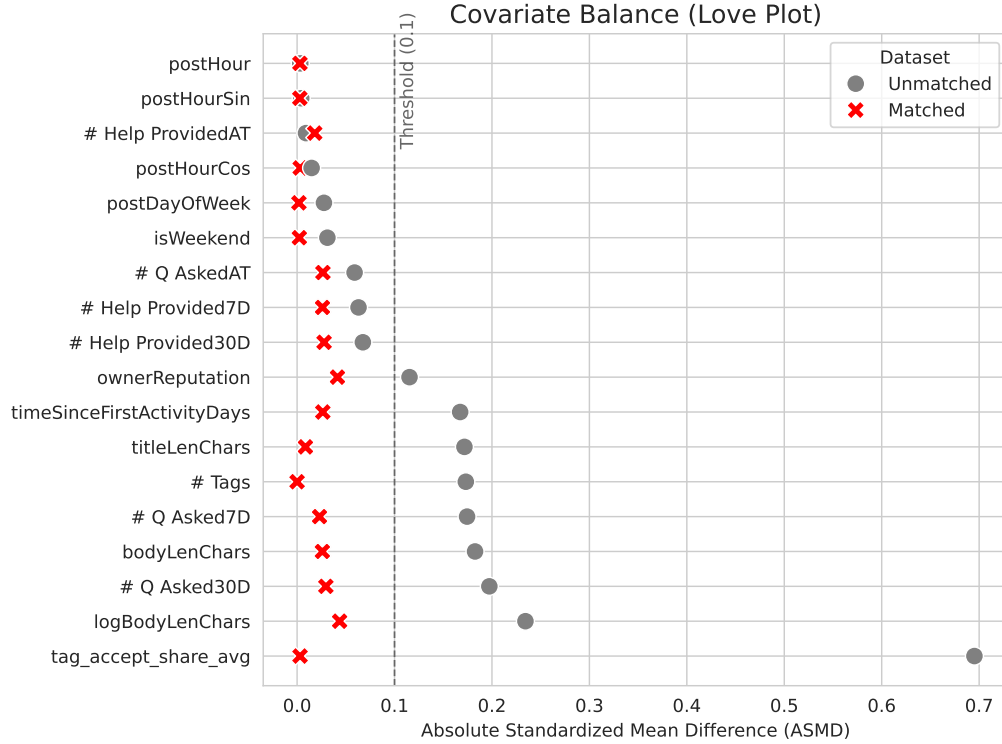


Figure B.8: Visualization of balance in the unmatched and matched dataset

Appendix C. Help Rate Over the Observation Window by Tenure

Figure C.9 shows that treated and control help rates track each other closely before the question in every tenure bucket, supporting the parallel-trends logic within each stratum. The sharp spike at question time is common to both groups and is somewhat larger for controls; the small post-answer elevations estimated in Table 7 and Figure 3 (hazard ratios of a few percent) are below what these raw normalized rates can visually resolve.

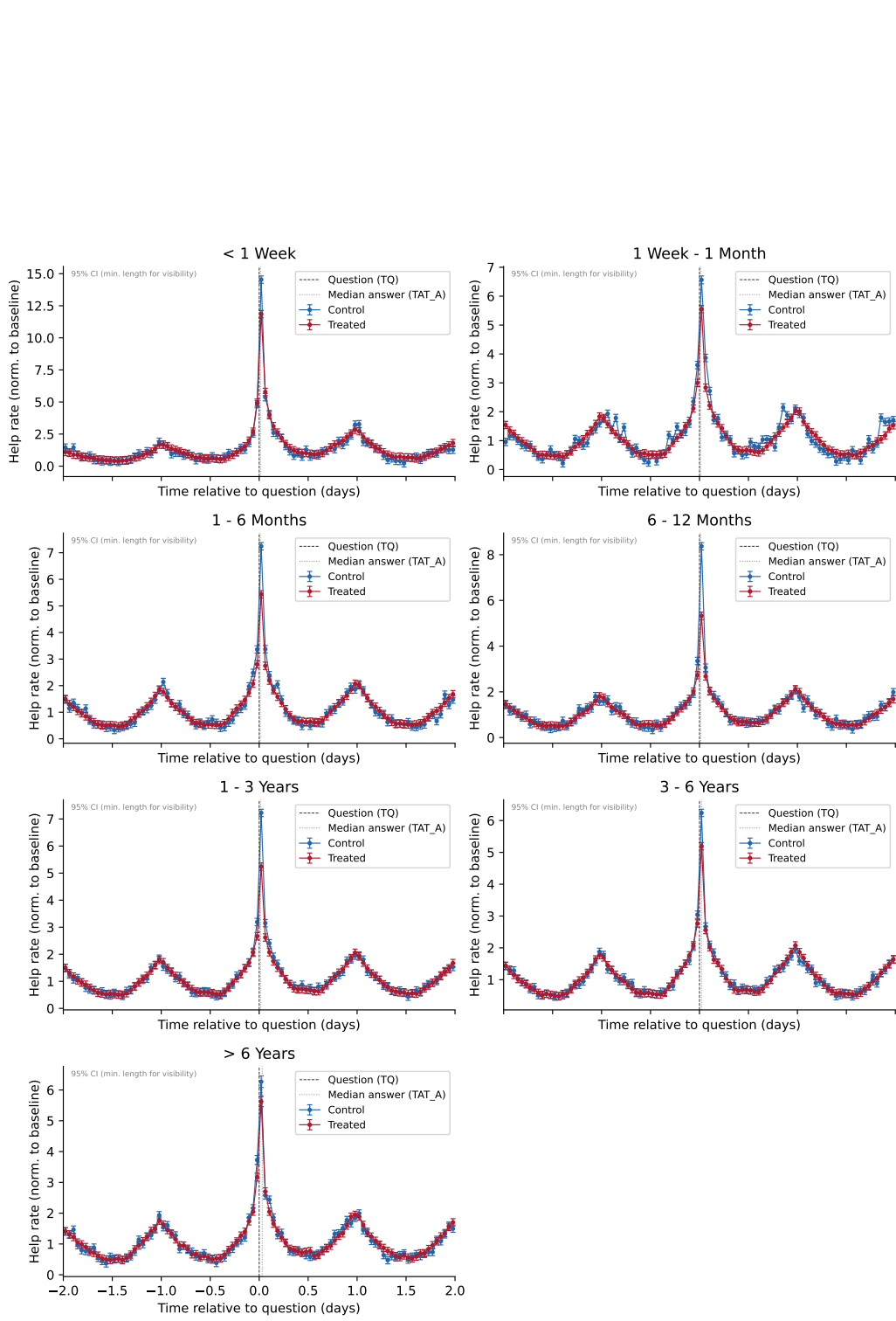


Figure C.9: Help Rate Over the Observation Window, by Tenure Bucket. Each panel plots the normalized help rate (relative to each group's pre-question baseline) for treated users (red) and matched control users (blue) over the ± 2 -day window around the question-posting time (T_Q , dashed vertical line). The dotted vertical line marks the median answer arrival time (T_A). Error bars indicate 95% confidence intervals.

Appendix D. Selection Sensitivity and Absolute Effects

Table D.11 reports transparent bounds on the reciprocity contrast under alternative assumptions about zero-event controls. Table D.12 translates the tenure-bucket hazard ratios into absolute-risk-difference proxies using observed baseline event rates. Table D.13 reports matched-pair bootstrap 95% confidence intervals for the pooled Model A treatment effect — the difference-in-differences increment at answer arrival (β_4) — based on 100 replicates.

Table D.11: Selection Sensitivity Based on Zero Post-Question Helping

Scope	Assumed frac.	Base HR (arrival)	Summed HR (upper bnd)	Control zero sha
All	0.00	1.00	1.01	93.9
All	0.10	1.00	1.01	93.9
All	0.25	1.00	1.01	93.9
All	0.50	1.00	1.01	93.9
< 1 Week	0.00	1.06	1.11	94.7
< 1 Week	0.10	1.06	1.11	94.7
< 1 Week	0.25	1.06	1.11	94.7
< 1 Week	0.50	1.06	1.11	94.7
1 Week - 1 Month	0.00	1.02	0.99	93.8
1 Week - 1 Month	0.10	1.02	0.99	93.8
1 Week - 1 Month	0.25	1.02	0.99	93.8
1 Week - 1 Month	0.50	1.02	0.99	93.8
1 - 6 Months	0.00	1.04	1.02	93.3
1 - 6 Months	0.10	1.04	1.02	93.3
1 - 6 Months	0.25	1.04	1.02	93.3
1 - 6 Months	0.50	1.04	1.02	93.3
6 - 12 Months	0.00	1.04	1.03	93.2
6 - 12 Months	0.10	1.04	1.03	93.2
6 - 12 Months	0.25	1.04	1.03	93.2
6 - 12 Months	0.50	1.04	1.03	93.2
1 - 3 Years	0.00	1.03	1.03	93.5
1 - 3 Years	0.10	1.03	1.03	93.5
1 - 3 Years	0.25	1.03	1.03	93.5
1 - 3 Years	0.50	1.03	1.03	93.5
3 - 6 Years	0.00	1.01	1.00	94.4
3 - 6 Years	0.10	1.01	1.00	94.4
3 - 6 Years	0.25	1.01	1.00	94.4
3 - 6 Years	0.50	1.01	1.00	94.4
> 6 Years	0.00	0.98	0.99	95.6
> 6 Years	0.10	0.98	0.99	95.6
> 6 Years	0.25	0.98	0.99	95.6
> 6 Years	0.50	0.98	0.99	95.6

Assumed frac. is the fraction of zero-post-help control questions assigned one latent help event.

Adjusted RR is an event-rate proxy, not a refitted Cox hazard ratio.

Base HR is the answer-arrival increment (β_4); Summed HR is the DiD $\exp(\beta_2 + \beta_4)$ (upper bound).

Table D.12: Absolute Effect Sizes and NNT

Row	Base risk	HR	ARD	NNT	NNT CI	Summed
< 1 Week	0.1181	1.059	0.00697	143.5	112.6–196.9	
1 Week - 1 Month	0.2001	1.021	0.00417	240.1	121.9–4868.1	
1 - 6 Months	0.2341	1.043	0.01001	99.9	75.3–147.5	
6 - 12 Months	0.2548	1.042	0.01073	93.2	70.4–136.9	
1 - 3 Years	0.2446	1.027	0.00664	150.5	101.7–286.5	
3 - 6 Years	0.1986	1.008	0.00155	645.2	—	
> 6 Years	0.1428	0.981	-0.00271	—	—	
AllData_Main	0.2064	1.002	0.00050	1995.0	—	

HR/ARD/NNT are the primary (headline) estimand; “Summed HR” is the summed DiD ($\exp(\beta_2 + \beta_4)$, upper bound).

Table D.13: Matched-Pair Bootstrap Uncertainty for the Summed DiD Treatment Effect (upper bound; Model A)

Scope	HR	Bootstrap 95% CI	Replicates
all	1.01	[1.01, 1.03]	100/100

Replicates resample whole matched pairs with replacement.

HR is the summed DiD ($\exp(\beta_2 + \beta_4)$, upper bound); the headline treatment effect elsewhere is the answer-arrival.

Appendix E. Robustness: Cohort Heterogeneity, Observable Selection, and Answer Quality

Table E.14 addresses the concern that pooling entry cohorts from 2008 through 2025 conflates structurally different eras of the platform (reputation inflation, tag evolution, and the post-2022 generative-AI disruption). Because calendar year enters the matching design only as an exact stratum, treated and control questions within a pair always share a year; to probe heterogeneity *across* years we re-estimate Model A on the pre-2020 subsample, both pooled and in the newcomer (< 1 week) stratum. The pooled pre-2020 arrival hazard ratio is `[[AO:pre2020-pooled-hr]]` (versus 1.03 in the full sample), and the newcomer association is `[[AO:pre2020-nc-pattern — essentially unchanged]]` (arrival HR = `[[AO:pre2020-nc-hr]]` pre-2020 versus `[[AO:nc-arrival-hr]]` in the full sample), indicating that the tenure gradient was already present before the generative-AI era.

Table E.14: Cohort Heterogeneity Robustness

Specification	Arrival HR	95% CI	Summed HR (upper bnd)	N	Events
ModelA_AllData	1.002	[0.989, 1.016]	1.008 [1.00, 1.02]	24,349,760	5,025,482
ModelA_Newcomer	1.059	[1.043, 1.075]	1.107 [1.09, 1.12]	4,029,008	475,142
ModelA_Pre2020	1.014	[1.001, 1.027]	1.013 [1.00, 1.02]	19,187,320	4,566,482
ModelA_Pre2020_Newcomer	1.056	[1.041, 1.072]	1.130 [1.12, 1.14]	3,155,176	432,142

Primary HR is the answer-arrival increment (β_4); the summed DiD ($\exp(\beta_2 + \beta_4)$) is shown as an upper bound.

Cox-table standard errors are model-based; matched-pair bootstrap uncertainty is reported separately when generated.

Events = helping events in the full analysis sample (after time rounding). When interval rows exceed the fit cap, estimation uses a matched-pair subsample, but Events still refer to the full sample.

Table E.15 reports the pooled Model A after adding pre-treatment observable selection controls (posting hour and weekday, number of tags, body and title length, and asker reputation). View count is *not* included: it accrues after posting and is higher for answered questions, so conditioning on it would open a post-treatment / collider channel already excluded from the propensity score. The pooled arrival increment remains [[AO:observables-pooled-pattern — essentially unchanged]] once these pre-treatment observables are held fixed (arrival HR = [[AO:observables-pooled-hr]], versus 1.03 in the baseline specification).

Table E.15: Observable Selection Controls

Specification	Arrival HR	95% CI	Summed HR (upper bnd)	N	Events
baseline	1.002	[0.989, 1.016]	1.008 [1.00, 1.02]	24,349,760	5,025,482
observable_controls	1.010	[0.999, 1.020]	1.012 [1.00, 1.02]	24,349,760	5,025,482

Primary HR is the answer-arrival increment (β_4); the summed DiD ($\exp(\beta_2 + \beta_4)$) is shown as an upper bound.

Cox-table standard errors are model-based; matched-pair bootstrap uncertainty is reported separately when generated.

Events = helping events in the full analysis sample (after time rounding). When interval rows exceed the fit cap, estimation uses a matched-pair subsample, but Events still refer to the full sample.

Table E.16 reports the newcomer (< 1 week) bucket under several specifications. The baseline arrival increment is [[AO:nc-arrival-hr]]. Adding

the same pre-treatment observable controls — still *without* view count — [[AO:nc-observables-pattern — verify: leaves the newcomer association positive (arrival X.XX)]], so the association is not an artifact of observable selection. Adding answer-quality controls, by contrast, [[AO:nc-quality-pattern — attenuates the newcomer increment markedly / pushes it below one]] (arrival [[AO:nc-quality-hr]]), the same pattern seen in the pooled effect and reinforcing that answer quality — not observable selection — is the most consequential confound the design does not resolve.

Table E.16: Newcomer Bucket Robustness (< 1 Week)

Specification	Arrival HR	95% CI	Summed HR (upper bnd)	
ModelA_Newcomer_Baseline (< 1 Week)	1.059	[1.043, 1.075]	1.107 [1.09, 1.12]	4,
ModelA_Newcomer_Observable (< 1 Week)	1.102	[1.089, 1.115]	1.217 [1.20, 1.23]	4,
ModelB_Newcomer_Baseline (< 1 Week)	0.987	[0.975, 0.999]	0.976 [0.97, 0.99]	4,
ModelB_Newcomer_LengthOnly (< 1 Week)	0.980	[0.968, 0.992]	0.961 [0.95, 0.97]	4,
ModelB_Newcomer_ScoreOnly (< 1 Week)	0.960	[0.948, 0.972]	0.923 [0.91, 0.93]	4,
ModelB_Newcomer_Quality (< 1 Week)	0.935	[0.924, 0.947]	0.877 [0.87, 0.89]	4,

Primary HR is the answer-arrival increment (β_4); the summed DiD ($\exp(\beta_2 + \beta_4)$) is shown as an upper bound.

Cox-table standard errors are model-based; matched-pair bootstrap uncertainty is reported separately when generated.

Events = helping events in the full analysis sample (after time rounding). When interval rows exceed the fit cap, estimation uses a matched-pair subsample, but Events still refer to the full sample.

Table E.17 revisits the discrete response-time bins after adding the same three answer-quality controls used in the pooled speed specification (acceptance, first-answer score, and length). Because each row already conditions on a response-time window, the quality-controlled fits use Model A covariates plus those three signals rather than continuous log-response-time interactions. Comparing columns shows how much of the bin-wise arrival pattern survives once observable quality is held fixed.

Table E.17: Treatment Effect by Response-Time Bin, With and Without Answer-Quality Controls

Response time	Baseline HR	95% CI	Quality HR	95% CI
0-15 min	0.99	[0.97, 1.01]	0.95***	[0.94, 0.97]
15-30 min	0.99	[0.96, 1.02]	0.95***	[0.93, 0.97]
30-60 min	1.22***	[1.19, 1.25]	1.06***	[1.04, 1.09]
1-2 hr	1.03*	[1.00, 1.06]	0.97*	[0.95, 0.99]
2-4 hr	1.03	[0.99, 1.06]	0.98†	[0.95, 1.00]
4-8 hr	1.00	[0.97, 1.03]	0.97†	[0.95, 1.00]
8-12 hr	0.97	[0.93, 1.01]	0.95*	[0.92, 0.99]
12-24 hr	1.04*	[1.00, 1.07]	1.02	[0.99, 1.05]
1-3 days	1.04**	[1.01, 1.08]	1.06***	[1.03, 1.09]
>3 days	1.01	[0.98, 1.05]	1.04*	[1.00, 1.07]

Cox-table standard errors are model-based; matched-pair bootstrap 95% CIs for the pooled DiD are in Table D.13.

Each row fits Model A to treated questions in that response-time bin plus the full no-answer control pool.

HR is the answer-arrival increment (β_4). Quality columns add acceptance, first-answer score, and length.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.1$

Appendix F. Implementation Details

All models are estimated using the `lifelines CoxTimeVaryingFitter` with a small ridge (L2) penalizer applied for numerical stability. The penalizer strength is $\lambda = 5 \times 10^{-3}$ for main-effect models and $\lambda = 10^{-2}$ for models that include response-time interaction terms (where the interaction covariates are correlated with the base treatment indicator). These values are sufficiently small that they introduce negligible shrinkage relative to the scale of the estimated effects, but they ensure stable Hessian inversion in large risk sets. All covariates are mean-centered before fitting; response-time interaction terms are additionally clipped to their 5th–95th percentile range and z-scored as described above. Cox-table standard errors are model-based (inverse-Hessian) and are *not* cluster-robust; dependence from matched pairs, reused controls, and repeated intervals can make those intervals too narrow. For the pooled Model A DiD we therefore also report a matched-pair bootstrap 95% CI (100 replicates resampling whole `match_id` pairs) in Table D.13. Ridge penalization and the winsorizing and standardization of the interaction covariates additionally regularize the reported model-based

standard errors. For the pooled experienced-user models (H1), point estimation uses a random subsample of approximately 8 million user-interval rows drawn from matched pairs (preserving pair integrity) when the full interval frame exceeds the fit cap; table *Events* counts still refer to the full analysis sample. Tenure-bucket-specific models (H2, H3) use all available intervals within each stratum.